Complete Vehicle Specification

VFDSXVEEP0047

Specification Book

Scope

Applicable Document of Part 1

Common Electrical Specification for Supplier

CES CAN

Range of validity

Vinfast

Document Administrator: Nguyen Dinh Dung

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Change Order not applicable **Status**: ISS (Inquiry Serial Production

Change History

Version	Brief Description	Date	Responsible person
1.0.0	First version of CAN CES base on the document ID: VFDSAAEEP0047 released V1.0.9	26.10.2019	Nguyen Dinh Dung
1.0.1	Added **Table 3-26: ECU behavior for Msg Rx & Tx** => as clarifications for supplier feedback on earlier version	24.08.2020	Rasul Nadaf
1.0.2	Mergred the B-CAR specification changes into this specification	21.09.2020	Rasul Nadaf
1.0.3	# Replaced the figure 3.12 as existing the image is not clearly visible # Updated table 3-28 for Autosar NM parameters with notes for partial networking	10.11.2020	Rasul Nadaf
1.0.4	# Deleted section 3.5.6.2- Timing parameters as this is redundant information to section 3.5.2.5.3 and 3.5.6.1 # Updated table 3-24 to specify timing parameter value for *N_Cs*	22.12.2020	Rasul Nadaf

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1 General Information

1.1 Language

This specification is written in English.

2 Introduction

2.1 Purpose of the document

The Controller Area Network (CAN) is a message-oriented Multi Master Bus System using CSMA/CR (Carrier Sense Multiple Access/Collision Resolution). For identification of separate messages, an unique Identifier (ID) with 11 Bit (according to CAN 2.0A), where the lowest ID represents the highest priority. Generally spoken, the CAN as a multi master system allows bus arbitration at all time for all ECU's with respect to their priority (message ID).

The NS 1 (ISO 11898-1) and NS 2 (ISO 11898-2) is based on the ISO/OSI model for layered data communication. This model outlines the seven distinct layers for general communication architecture in which the definitions of the physical layer of CAN are separated in two specifications. NS 2 (ISO 11898-2) specifies the requirements of high-speed CAN for transmission rates is 500 kBit/s. The specifications about the physical layer also contain the requirements to the physical media. In general, for the connection of all CAN nodes a two-wire differential bus is used.
The data link layer is described NS 1 (ISO 11898-1). This specification defines the physical signaling of CAN as well as the associated mechanism of medium access control. In general, this corresponds to implementation of CAN-controller on MCUs.

2.2 Terms and definitions

This chapter provides the details about Terms and definitions.

2.2.1 Abbreviations

The following list of abbreviations are used throughout this document with the given meaning.

2.2.1.1 Temperatures

Tmin	Minimum operating temperature
Tamb	Ambient temperature
Tmax	Maximum operating temperature
TTest	Test temperature

Table 2-1: Temperatures

2.2.1.2 Voltage and Current

UBmin	Lower operation voltage limit
UB	Operation voltage
UBmax	Upper operation voltage limit

Umax	Maximum voltage during test
Umin	Minimum voltage during test
UPP	Peak-peak voltage
Ueff	Root-mean-square value of the voltage
UTest	Testing voltage
IN	Nominal current
GND	Device ground
DUT	Device under test
UA, UT, US, UR	Voltage levels during cranking profile

Table 2-2: Voltage and Current

All specified voltages and currents are related directly to the DUT. Additional voltage drops at the energy harness are not included if not mentioned separately.

2.2.1.3 Timing

tr	Rise time
tf	Fall time
tTest	Test duration

Table 2-3: Timing

2.2.1.4 Other

ACK	Acknowledgement
ACC	Clamp ACC or Radio
AUTOSAR	Automotive Open System Architecture
API	Application Program Interface
BNDB	Board Net Data Base
BR	Bit Rate
CAN	Controller Area Network
CCP	CAN Calibration Protocol
CL15	Clamp 15 (Ignition state/line) equal to ignition=ON
CL30	Clamp 30
CL30S	Clamp 30, switched
CL50	Clamp 50, switched, turning the starter on
CRC	Cyclic Redundancy Check
CSMA/CA	Carrier Sense Multiple Access/Collision Avoidance
CSMA/CR	Carrier Sense Multiple Access/Collision Resolution
DLC	Data Length Code
DTC	Diagnostic Trouble Code
ECU	Electronic Control Unit
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EOF	End of Frame
HS-CAN	High Speed CAN

IDE	Identifier Extension Flag
ID	Identifier
ISO	International Standardization Organization
LS-CAN	Low Speed CAN
LSB	Least Significant Bit
MCU	Micro Controller Unit
MSB	Most Significant Bit
NM	Network Management
NA	Not Available
NMPDU	Network Management Protocol Data Unit
OBD	On-Board Diagnosis
OSEK	Open Systems and the Corresponding Interfaces for Automotive Electronics
OSI	Open System Interconnection
PDU	Protocol Data Unit
PLL	Phase Locked Loop
PN	Partial Network
REC	Receive Error Count
RMR	Repeat Message Request
RTR	Remote Transmission Request
ROM	Read-Only Memory
OEM	Original Equipment Manufacturer
SAE	Society of Automotive Engineers
SOF	Start of Frame
SP	Sample Point
SJW	Resynchronization Jump Width
St_{min}	Minimum Separation Time
TBD/tbd	to be defined
TA	Target Address
TEC	Transmit Error Count
TTC	Time Triggered Communication

Table 2-4: Abbreviations

2.2.2 Definition

The following list of definitions are used throughout this document with the given meaning.

2.2.2.1 Obligation of Requirement Wording

The following wording rules are used throughout this document.

Shall

The word "shall" is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

Should

The word "should" is used for preferences or desires in implementation that are not strictly binding.

May

The word "may" is used to denote options that are left to the implementer.

2.2.2.2 Standard Tolerances

Frequency	±1%
Temperature	±2°C
Humidity	±5%
Times	+5%; -0%
Voltages	±2%
Currents	±2%

Table 2-5: Standard and Tolerances

2.2.2.3 Standard Values

Room temperature	TRT=23°C
Humidity	Frel=45%-75% RH
Test temperature	TTest = TRT
Nominal voltage	UN=12V
Operation voltage	UB=14V

Table 2-6: Standard Values

2.2.2.4 Other

The following list of terms and definitions are used throughout this document with the given meaning.

bit rate

The number of bits transmitted per time, independent of bit representation [QD1].

bit stuffing and **destuffing**

Insertion and removal of fill-bits to provide bus state changes after a specific number of equal bits. This is required for periodic resynchronization when using a non-return-to-zero bit representation [QD1].

bus

A network topology in which all nodes are interconnected with each other using passive links that allow transmission in both directions [QD1].

dominant and recessive

The two states of the physical communication line are called dominant and recessive. The recessive state can be overwritten by the dominant state but not vice versa.

frame

A data unit used by the data link layer for transmission across the transfer medium with a specified position and meaning for each data bit [QD1].

node

An assembly that is connected to a communication network. Nodes are capable of communicating across the network following some communication protocol specification [QD1].

non-return-to-zero

Characteristic of binary signals where the signal level within a single bit time does not change and where no edges occur within a sequence of bits having equal signal levels.

protocol

A formal set of rules that define the information interchange between nodes [QD1].

receiver

A node is a receiver if it is not transmitting and if the bus is not idle [QD1].

transmitter

A node sending a frame onto the bus. It stays transmitter until the bus is idle again or until it has lost arbitration [QD1].

2.3 CAN and the OSI-reference Model

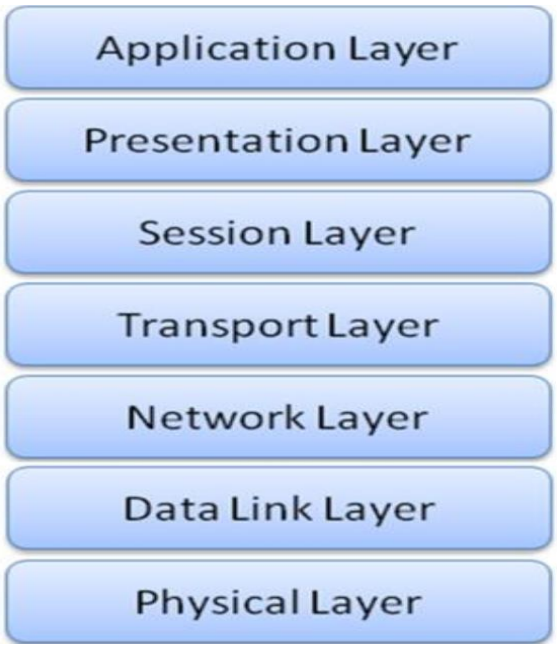


Figure 2-1 OSI network Model

Figure OSI Network Model outlines the seven distinct layers for general communication architecture in which the layers shall be understand as described in section System description.
The aim of this document is to define all necessary requirements for the development of a CAN bus system used in the road vehicles.

3 System Description

3.1 Specific Terms and Definitions

The following list of terms and definitions are used throughout this document with the given meaning.

3.2 Overview

The bus arbitration for all ECU's takes place BIT-by-BIT and, as already mentioned, the highest priority message receives unimpeded access to the bus, not the matter of how many ECUs require bus access. All nodes (e.g. ECUs) are allowed to listen to the bus at all time. That implies that every node can receive every message sent on the bus.

Sending a message is possible as soon as the bus is free, starting with the arbitration process. As already mentioned, each CAN message is unique (via ID) and can only be sent by one node.

Furthermore, at least one node must send an acknowledge bit as a hint that the message has been successfully received (transmitted). Otherwise the transmitting node keeps up sending the same message again and again or report an acknowledge error to ECU-application, which initiated message transmission.

3.3 Physical Layer

3.3.1 Introduction

The purpose of this section is to define requirements of physical signaling on a CAN bus system for road vehicles. More precisely, the section specifies the physical layer following the OSI (Open System Interconnection) model for a Carrier Sense Multiple Access/Collision Resolution(CSMA/CR) interconnection for communication among electronic control units (ECU) on road vehicles. Therefore, this document is to be referenced by every ECU specification that incorporates interconnection for ECU communication using CAN. The ISO 11898 CAN specifications (NS 1, NS 2, NS3,) are based on the ISO/OSI model for layered data communication. Figure ISO/OSI model outlines the seven distinct layers for communication. In this document the first layer for CAN communication is specified based on ISO standards and SAE recommendations. The physical layer describes the electrical means and their logical interpretation for communication.

3.3.2 High-Speed-CAN

3.3.2.1 BUS Topology

Figure 3-1 shows the bus topology that shall be used and star topology is not allowed. ECU1 and ECUn at the end of the bus topology shall have an internal bus termination. ECU1 and ECUn are typically located at the largest bus cable distance from each other. All other ECUs are connected by a stub. To minimize standing waves, ECUs should not be placed equally spaced on the network

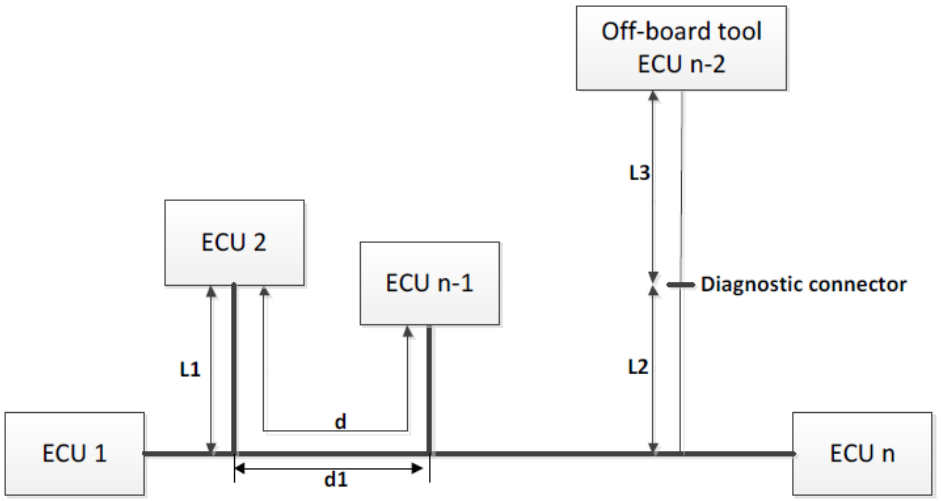


Figure 3-1 Bus Topology

The detailed parameters see Table 3-1.

Parameter	Symbol	Unit	Min	Nominal	Max	Comments
ECU Distance	d	m	0.1	-	30	Distance between any two ECUs on the bus, including cable stubs and an Off-Board Tool.
Cable stub length	d1	m	0.1	-	-	Exception：If d1 is zero, it can be only connect two ECUs at the point.
ECU Cable Stub Length	L1	m	0	-	1	Any ECU's L1 should be unequal length with others.
Cable Stub Length off-board tool	L2	m	0	-	1	
Maximum stub length off board tool	L3	m	0	-	5	

Table 3-1 Network Topology Parameters

3.3.2.2 Wiring

The wires for CAN high should have 0.35 mm² wire gauge.

For further specifications refer to NS 1 and NS 2 (ISO 11898-1 and ISO 11898-2).

The requirements of the wiring see Table 3-2.

Parameter	Symbol	Min	Norm	Max	Unit	Conditions
Impedance	Z	90	120	140	Ω	f=1 MHz
Length Related Resistance	R_{Length}			120	mΩ/m	Single conductor
Specific Line Delay	T_{Delay}			5.5	ns/m	Wire only
Rate of Twist	RateTwist	33	40	50	Twist/m	360 degrees

Table 3-2 Physical Media Parameters for Unshielded Twisted Pairs

3.3.2.3 Bus interface HS_CAN

Figure 3-2 provides a recommended schematic that shows how the bus interface shall be implemented. The layout guidelines from the transceiver manufacturer shall be considered. It shall be ensured that the CAN transceiver is placed near the ECU connector due to EMC reasons.

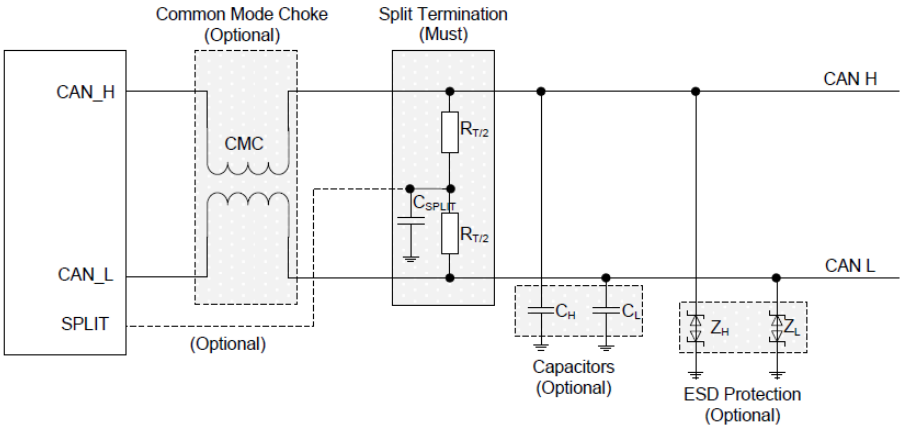


Figure 3-2 High Speed CAN Bus Interface

3.3.2.4 High-speed-CAN Split Termination

Capability to support termination shall be provided on the PCB by each ECU. The ECUs that will populate the termination network will be defined specifically for each vehicle program.
The termination resistors should be metal film resistors with a tolerance of 1% and a power dissipation of 1/3W. For bus termination two resistors with 60 Ohm shall be used (split termination). Other variants are not allowed. The termination resistors shall be included in ECUs at the beginning and the end of the bus system.

The termination components see Table 3-3.

Parameter	Symbol	Unit	Min	Nom	Max	Comments
Resistor 1)	$R_{T/2}$	Ω	59	62	66	Min. power dissipation: 110mW Tolerance shall be ±1%
Resistor 2)	$R_{T/2}$	kΩ	1.3		4.7	
Capacitor 1)	C_{SPLIT}	nF		4.7		
Capacitor 2)	C_{SPLIT}	nF	4.7		100	

Table 3-3 Termination Component

Note: 1) For terminal nodes
2) For other nodes, the footprints for bus terminal components on the PCB shall be reserved.

3.3.2.5 High-speed-CAN physical signaling

The physical signaling shall comply to the requirements in [NS2] and [NS3] (ISO 11898-2 and ISO 11898-3). The rising and falling time (slew rate) must be shorter than the times issued in Figure 3.3:

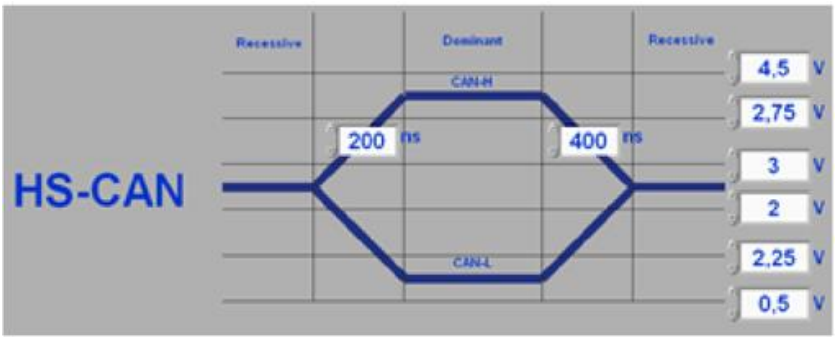


Figure 3-3 HS-CAN signal levels and slew rates

Parameter	Symbol	Unit	Value			Condition
			Min	Nom	Max	
Output voltage	V_{CAN_H}	V	2.0	2.5	3.0	No Load
	V_{CAN_L}	V	2.0	2.5	3.0	
Differential Output Bus Voltage	V_{diff}	mV	-50	0	50	No Load

Table 3-4 Output Bus Voltage Parameter for Recessive State

Parameter	Symbol	Unit	Value			Condition
			Min	Nom	Max	
Output voltage	V_{CAN_H}	V	2.75	3.5	4.5	Load RT/2[1)]
	V_{CAN_L}	V	0.5	1.5	2.25	
Differential Output Bus Voltage	V_{diff}	V	1.5	2.0	3.0	Load RT/2[1)]
Note 1): The load is connected between CAN_H and CAN_L. For a CAN node without integrated terminating resistor (normal use), this resistor is a $R_{T/2}$ resistor. For CAN nodes with an integrated terminating resistor, this is a $R_{T/2}$ x 2 resistors						

Table 3-5 Output Bus Voltage Parameter for Dominant State

3.3.2.6 EMC Aspects

It shall be ensured that the CAN-transceiver is placed near the ECU-connector due to EMC reasons.

3.3.2.7 Common mode choke

To reduce EMC disturbances, it may be necessary to use a common mode choke (CMC). Therefore, the PCB layout shall accommodate footprints for placing a bus EMC inductor (15 ~ 33µH, e.g. EPCOS B82789C0223). The footprint provision is also required when all EMC tests have been successfully passed without inductor. If a bus inductor was not necessary for a certain component, then the footprints shall be populated with zero Ohm resistors or a conductive path between the footprints.

3.3.2.8 Capacitors

Between CAN-H and GND and between CAN-L and GND a capacitor with 47pF/50V shall be populated. If the CAN-transceiver is not optimally placed near the connector it may be useful to populate two capacitors on the PCB. One near the connector and one near the CAN-transceiver. It shall be ensured that the overall capacity is not greater than 47pF. The actual capacitance value of both parts shall be close to each other to optimize EMC behaviors. If a special ESD protection circuit is used, then the capacitors might have to be chosen with a lower value or be omitted, depending on the capacitance of the protection circuit. Both capacitors shall be placed close to the bus connector with minimum trace length.

Parameter	Symbol	Unit	Min	Nom	Max
Capacitor	C_H, C_L	pF			47

Table 3-6 Capacitors

3.3.2.9 ESD-Protection

Each ECU shall meet the applicable ESD requirements at the CAN bus input pins. This includes ESD

during operation and ESD protection for packaging & handling. The ESD transient suppressor technique shall be compliant to the common mode bus voltage range. Therefore, the ESD protection circuit shall exhibit a high impedance in this voltage range, e.g.: Rin_ESD > 100 kOhm for -7V < UCAN_H/L < +12V. The ESD protection circuit shall not degrade the bus input resistance and bus input capacitance matching of the CAN interface. Possible ESD protection components include e.g. back-to-back Zener diodes, MLE, MOV or other suppression parts that do not exceed the capacitance limits specified in NS 3 (ISO 11898-3). The ESD transient suppressor components should provide a breakdown voltage which is higher than the maximum battery voltage at jump start condition.

3.3.2.10 External ESD-Protection

External ESD-Protection is not allowed on CAN-Bus-systems. Deviations must be defined together with the functional department.

3.3.2.11 Bus Interface with Wake-Up Function

All ECU´s within the High-speed-CAN network shall have implemented CAN-transceiver which are able to be waked up by a CAN-message (exceptions for carry over parts possible). Transceivers which support the wake-up patterns according to [NS1] and [NS2] (ISO 11898-1 and ISO 11898-2) should be used. Additional information is included in the document "CES_Clamp-Specification".

Deviations have to be agreed by OEM.

3.3.2.12 Controller Oscillator

The oscillator provides the reference timing for the CAN controller. The frequency of the oscillator shall be an integer multiple of the baud rate. PLL should not be used for the reference timing, because the processing load of the microcontroller can produce a jitter. Only crystals shall be used for clock generation. Other clock generators, e.g., ceramic resonators are not allowed. The tolerance of the crystals shall be less than ± 0.15% over the according temperature and aging ranges.

3.4 Data Link Layer

3.4.1 Introduction

The purpose of this section is to define requirements of data transmission on a CAN bus system for road vehicles more precisely. This section specifies the Data Link Layer following the OSI (Open System Interconnection) model for a Carrier Sense Multiple Access/Collision Resolution (CSMA/CR) interconnection for communication among electronic control units (ECU) on road vehicles. Therefore, this section is to be referenced by every ECU specification that incorporates interconnection for ECU communication using CAN. ISO 11898 CAN specifications [NS1], [NS2] (ISO 11898-1, ISO 11898-2) are based on the ISO/OSI model for layered data communication. The figure of the ISO/OSI model outlines the seven distinct layers for communication. In this section the Data Link Layer is specified based on ISO standards and SAE recommendations.

3.4.2 Message format

3.4.2.1 High speed CAN (HS-CAN) structure

SOF (1 bit)	Identifier (11 bits)	RTR (1 bit)	IDE (1 bit)	R0 (1 bit)	DLC (4 bits)	Data (0 – 64 bits)	CRC (15 bits)	CRC delimiter (1 bit)	ACK (1 bit)	ACK Delimiter (1 bit)	EOF (7 bits)	IFS (3 bits)

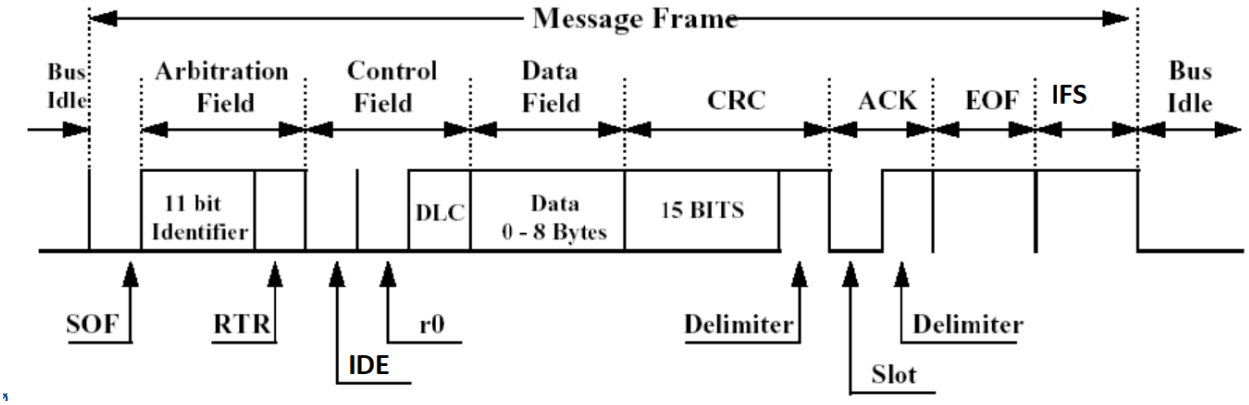


Figure 3-4 Message frame structure in HS-CAN

Field name	Length (bits)	Purpose
SOF	1	The single dominant start of frame (SOF) bit marks the start of a message, and is used to synchronize the nodes on a bus after being idle. [Dominant (0)]
Identifier	11	The Standard CAN 11-bit identifier establishes the priority of the message. The lower the binary value, the higher its priority.
RTR	1	The single remote transmission request (RTR) bit is dominant when information is required from another node. Least significant bit of arbitration field [Dominant (0)]
IDE	1	Identifier Extension Dominant (0) for standard frame format, Recessive (1) indicates extended frame format
Reserved (r_0)	1	Reserved bit (for possible use by future standard amendment) [Dominant (0)]
DLC	4	Data length code Number of bytes of data (0–8 bytes) C[3] – C[0]: length of data array, MSB first.
Data	0 to 64	Data to be transmitted (length in bytes dictated by DLC field)
CRC	15	The 15-bit cyclic redundancy check (CRC) contains the checksum (number of bits transmitted) of the preceding application data for error detection.
CRC delimiter	1	CRC delimiter Must be recessive (1)
ACK	1	ACK is 2 bits, one is the acknowledgment bit and the second is a delimiter.

		Every node receiving an accurate message overwrites this recessive bit in the original message with a dominate bit, indicating an error-free message has been sent. Should a receiving node detect an error and leave this bit recessive, it discards the message and the sending node repeats the message after re-arbitration. In this way, each node acknowledges (ACK) the integrity of its data.
ACK delimiter	1	ACK delimiter Must be recessive (1)
EOF	7	This end-of-frame (EOF), 7-bit field marks the end of a CAN frame (message) and disables bit stuffing, indicating a stuffing error when dominant. When 5 bits of the same logic level occur in succession during normal operation, a bit of the opposite logic level is stuffed into the data.
IFS	3	This 3-bit Inter Frame Space (IFS) contains the time required by the controller to move a correctly received frame to its proper position in a message buffer area.

Table 3-7 Description of CAN message frame's fields

3.4.3 General Design Requirement

More detailed specification on data link layer can be found in [NS1] (ISO 11898-1).

CANDLL_GENERAL_001:

The standard message format with 11-bits identifier shall be used.

CANDLL_GENERAL_002:

The extended message format with 29-bits identifier shall not be used.

CANDLL_GENERAL_003:

Remote frame and overload frame as specified in [NS1] (ISO 11898-1) shall not be used.

3.4.4 Baud Rate and Bit Timing Parameters

HS-CAN operates with 500kbit/s. Telematics, Infotainment, safety, Powertrain, Body and Diagnostics connected to the HS-CAN. In general, the bit timing parameters (see Figure example of bit timing parameters for HS-CAN with 4MHz oscillator frequency.) for each CAN controller shall be configurable according to specification in [NS1] (ISO 11898-1).

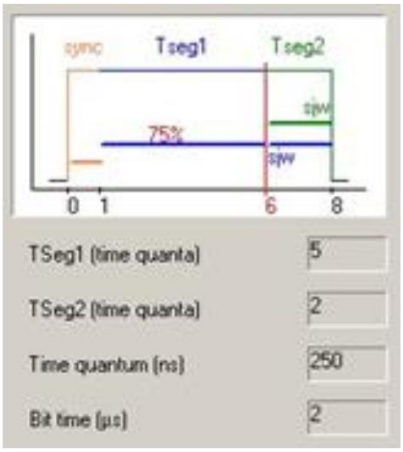


Figure 3-5 Bit Timing Parameters for HS-CAN with 4 MHz Oscillator Frequency

High speed CAN (HS-CAN):

	Oscillator frequency				Unit
	8	**10**	**12**	**16**	**MHz**
Baud rate	**500**				**kBaud**
1 Bit time (bt)	**2**				**µs**
1 Time quantum (qt)	**1/8 bt**	**1/10 bt**	**1/12 bt**	**1/16 bt**	
Synchronization jump width	**1**			**1 or 2**	**qt**
Sample	**1**				
Sample point	**65-85**				**%**
Resynchronization	At the recessive to the dominant edge				

Table 3-8 Recommended Bit Timing Parameter Configuration for HS-CAN

Individual or special configuration for ECUs (whether in sample status or series status) must be coordinated with OEM, if the recommended configuration not possible for some reasons like EMC or safety- requirements.

3.4.5 Communication Error Detection and Recovery

A CAN node shall be able to distinguish between short disturbances on the bus and permanent failures. A node that is transmitting erroneous data permanently shall be switched off. A switched off node is logically disconnected from the bus such that it can neither transmit nor receive any data. According to NS 1 (ISO 11898-1), the objective of a fault confinement is to preserve a high availability of the data transmission system, even in the case of a node defect. Therefore, the fault confinement strategies have to prove reliable on the distinction between temporary errors and permanent failures, and localization and switching-off the faulty node. Based on error counter, the CAN controllers implement Bus off error detection as defined in [NS1] (ISO 11898-1). The state change from bus-off to error-active must be enforced by a reset. After a reset the node shall not start transmission of frames until 128 occurrences of 11 consecutive recessive bits are observed on the bus. This allows all other nodes to successfully send at least 128 frames without being disturbed by a permanently faulty node.

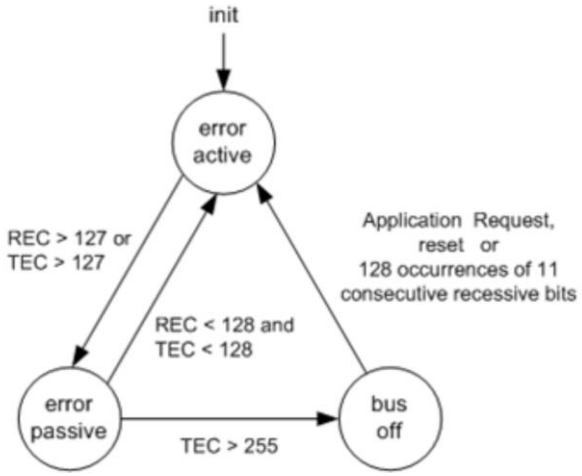


Figure 3-6 Node Status Transition Diagram in CAN-Controller

3.4.6 Node States and Status provided by CAN-Controller

Fault confinement is based on three states with different behavior as described in [NS1] (ISO 11898-1). The status of this states, the status information of CAN-controller and transceiver shall be providing from CAN-Driver. In detail following information, shall be provide by CAN-driver software:

- Sleep status, indicating if CAN Controller and transceiver are switch to sleep/standby mode usually called power-down mode to reduced power consumption
- Status of CAN Controller is in stop mode, indicating if CAN-controller and transceiver are initialized but stopped. No message transmission or reception is possible.
- Offline status mode, indicating if CAN Driver transmit path (controller and transceiver) is disabled. No transmission possible but reception may be possible.
- Error states of the CAN Controller (Error-Active, Warning, Error-Passive or Buss-Off) as specified in [NS1] (ISO 11898-1).

3.4.7 Bus-Off

For prevention of permanent disturbance of the bus communication due to send problems, every ECU shall be able to detect bus-off by using information provided by CAN-controller. The CAN Driver shall provide functions to handle a bus-off state of the CAN Controller after severe transmit errors. For some CAN Controllers a re-initialization must be done to satisfy the hardware requirements. For any controllers, this changes automatically to the 'Error Active' state after 128 x 11 recessive bits on the CAN bus as it is specified in the CAN protocol. Nevertheless, it is recommended by most of the customer specific CAN bus specifications to re-initialize the CAN Controller in every case, because the transmit error might be caused by a faulty bit in the CAN Controller registers, e.g. bus timing registers, in case of EMC influences. A node in bus-off state is switched off and the following behavior shall be performed:

- In the bus-off state, a node shall not send any frames.
- A node shall start the recovery from bus-off state only upon ECU-application request.

3.4.8 Recovery from Bus-off

If a node enters bus-off it shall immediately stop message transmissions for configurable amount of time (BUSOFF_RECOVERY_TIMEOUT) but it may continue to receive messages. After

BUSOFF_RECOVERY_TIMEOUT the node shall re-initialize the CAN-controller and restart normal message transmission. If the node reenters bus-off it shall perform this procedure indefinitely until normal operation conditions are restored.

$$\boldsymbol{BUSOFF_RECOVERY_TIMEOUT = 1000ms}$$

3.5 Network Layer

3.5.1 Introduction

This section specifies the Network Layer for communication among electronic control units (ECU) on road vehicles and is referenced by every ECU specification that incorporates interconnection for ECU communication. In this section the Network layers for CAN communication (see ISO/OSI - Model) is specified based on ISO standards and SAE recommendations.

3.5.2 Network Layer Frames

The network layer (transport protocol) uses the CAN data frames as described in following tables. The data length code (DLC) of the used CAN-frame shall be always set to 8 according to NS 6 (ISO 15765-2). If the number of protocol data bytes to be transmitted is shorter than 8 bytes, then the sender has to set the DLC to the maximum value 8 (padding of unused data bytes). In particular, this can be the case for a single frame (SF), flow control frame (FC) or last consecutive frame (CF) of a segmented message. The DLC parameter of the CAN frame is set by the sender and read by the receiver to determine the number of data bytes per CAN frame to be processed. This DLC of CAN-frame cannot be used to determine the length of protocol data bytes; this information shall be extracted from the protocol control information (**PCI**) in the beginning of protocol data.

Address information	PCI	Data field
11 bit	1 to 3 bytes (3 bytes for FC)	6 to 7 bytes
CAN Identifiers	See section: Protocol control information (PCI)	Protocol data

Table 3-9 Structure of data frames used by network layer

11 Bits CAN-ID	DLC	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
BA + SA	**8**	**TA**	Protocol data **(2 … 7 Bytes)**						

Table 3-10 TP Data in CAN-Frame using 11 bits "extended addressing"

11 Bits CAN-ID	DLC	Byte1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
BA + SA or BA+ TA	**8**	Protocol data **(2 … 8 Bytes)**							

Table 3-11 TP Data in CAN-Frame using 11 bits "normal addressing

BA = Base Address (e.g. 0x600) for diagnostic- or TP-messages are defined in [QD2]
SA = Source Address defined in [QD2]
TA = Target Address defined in [QD2]

The transport protocol specified in the network layer defines 4 different frame types as described in following sections Single Frame to Flow Control Frame. More information and details specification can be founded in NS 6 (ISO 15765-2).

3.5.2.1 Protocol control information (PCI)

Frame structure of the PCI, see Table: Protocol control information using 11 bits "normal addressing". For Protocol control information using 11 bits "extended addressing". For more detailed description of the PDU types see ISO 15765-2 [NS6].

PDU type	PCI bytes			
	Byte 0		Byte 1	Byte 2
	Bits 7-4	Bits 3-0	all	all
Single frame (SF)	0x0	Data length		
First frame (FF)	0x1	MSB (Bit3)	LSB (Bit0)	
		Data length 12 bits		
Consecutive frame (CF)	0x2	Sequence number (SN)		
Flow control (FC)	0x3	Flow status (FS)	Block Size (BS)	STmin

Table 3-12 Protocol control information using 11 bits "normal addressing"

3.5.2.2 Single Frame

The single frame (SF) is used to send protocol data with a size less than 7 bytes. In this case all data bytes fit into a single CAN frame of the data link layer and no segmentation is required. The single frame format is depicted in Table: Single frame format. The first byte contains the protocol control identifier (PCI) type 0x0 and the single frame data length (SF_DL) which denote the number of data bytes that are contained in the protocol data. If the number of data bytes is less than 7 (1 PCI-byte + 6 data bytes), padding is used to fill all 8 data bytes of the CAN message.

Byte#1		Byte#2	Byte#3	Byte#4-8
Bit 7-4	Bit 3-0			
PCI Type 0	SF_DL	Data Padding		

Table 3-13 Single Frame Format using "normal addressing"

3.5.2.3 First Frame

The first frame (FF) is used to indicate the start of a data transmission that consists of more than 7 bytes. With the first frame the receiver is informed about the complete number of data bytes (FF_DL) that will

be transmitted. In addition, the first 5 bytes are included in the first frame. The PCx1 for first frames as can be seen in Table: First frame format.

Byte#1		Byte#2	Byte#3	Byte#4-8
Bit 7-4	Bit 3-0			
PCI Type 1	FF_DL		Data	

Table 3-14 First Frame Format using "normal addressing"

3.5.2.4 Consecutive Frame

Consecutive frames (CF) is used to transmit follow-up data bytes. The PCI type for consecutive frames is set to 0x2. This frame also contains a sequence number that allows the receiver to detect missing frames. The sequence number is incremented on every consecutive frame starting at 1 (sequence number 0 is assigned to the corresponding first frame). If the sequence number reaches 15 (0xF) the next consecutive frame shall get a sequence number of 0 (0x0). Table:
Consecutive Frame Format shows that the 6 protocol data bytes of the consecutive frame may contain padding bytes. These padding bytes are only used for the last consecutive frame.

Byte#1		Byte#2	Byte#3	Byte#4-8
Bit 7-4	Bit 3-0			
PCI Type 2	SN	Data Padding		

Table 3-15 Consecutive Frame Format using "normal addressing"

3.5.2.4.1 Sequence number (SN)

Used for the CF, it specifies the frame number transmitted, the allowed values from 0x0 to 0xF, when reaching the 0xF value in begins again with 0x0. For detailed information see content of [NS6].

3.5.2.5 Flow Control Frame

With flow control (FC) frames the receiver indicates its state to the transmitter. A receiver of a first frame uses flow control to acknowledge the reception and to request the next block of continuous frames. The PCI type of flow control frames is set to 0x3. As it can be seen in Table: Flow control frame format, flow control frames contain a flow status field (FS), a block size field (BS), and a minimum separation time field (STmin).
The flow control frame shall always be sent after a First frame (FF). For transmissions with BS = 0 no more FC has to be sent.

Byte#1		Byte#2	Byte#3	Byte#4-8
Bit 7-4	Bit 3-0			
PCI Type 3	FS	Block Size (BS)	STmin	N/A

Table 3-16 Flow Control Frame Format using "normal addressing"

Signification and description of the flow control states values (FS) are listed in Table Definition of flow state values.

Value	Description
0	Continue to send: the sending node shall continue to send consecutive frames.
1	Wait: the sending node shall wait for another flow control frame and shall restart its timeout counters.
2	Overflow: the sending node shall immediately abort transmission of segmented messages. This value is only allowed in the flow control message immediately after the reception of a first frame message and shall be used to indicate that the message length FF_DL exceeds the buffer size of the receiving node.
3-F	Reserved: these values shall be reserved

Table 3-17 Definition of Flow State Values

With the block size field, the receiver tells the data transmitter to limit the number of unconfirmed consecutive frames. A value between 0x01 and 0xFF cause the transmitting node to wait for a flow control frame after sending block size number of consecutive frames. A block size of 0 causes the transmitting node to send all consecutive frames without waiting for any further flow control frames.
The STmin-Value indicates the transmitting node to insert a delay (minimum separation time) between the transmission of consecutive frames. Table: Definition of minimum separation time depicts the minimum separation time that shall be guaranteed (see ISO 15765-2).

Value	Description
00 - 7F	**Separation Time range: 0 ms - 127ms** STmin values from 00 hex to 7F hex denote absolute milliseconds (ms).
80 - F0	**Reserved**: Values in this range are reserved.
F1 - F9	**Separation Time range: 100 µs - 900 µs** The values in the range F1 hex to F9 hex denote even 100 microseconds (µs) starting at F1 hex
FA - FF	**Reserved**: Values in this range are reserved.

Table 3-18 Definition of Minimum Separation Time

3.5.2.5.1 Flow status (FS)

The flow status depicts table: Flow status (FS)controls the data flow for the transmission for detailed information, see content of document [NS6]. All devices are always ready to send after the TP channel has been established. The wait function shall not be used. Therefore, the FS (First Frame) value is always set to 0x00.
FS = 0x00
An implementation with the wait function could be possible at a later date, if the interaction with the tester

requires this feature.

Hex Value	Description
0x0	Continue To Send (CTS)
0x1	Wait (WT)

Table 3-19 Flow status (FS)

3.5.2.5.2 Block Size (BS)

The block size indicates the number of data frames that could be transmitted without expecting a flow control. Table: BS-values specified by OEM below give the BS-values specified by OEM.

Hex Value	Description
0x0	No more flow controls (FC) to send
0x1-0xFF	Maximum number of CF without FC

Table 3-20 Block Size

Parameter	OEM enhanced diagnostics	OBD diagnostic	ECU Programming (flashing1)	User data
Block size (BS)	8	0	0	8

Table 3-21 BS-values specified by OEM

3.5.2.5.3 ST_{min}

The STmin indicates the minimum time slot between two consecutive frames.
If STmin = 0 the CF's will be sent as fast as possible. Other values could be necessary if the transmission with FS is used.

Hex Value	Description
0x00 – 0x7F	Separation time in milliseconds. Range: 0ms – 127ms. This time value is specified in chapter Network layer parameters

Table 3-22 STmin-range

Parameter	OEM enhanced diagnostics	OBD diagnostic	ECU-re Programming (flashing)1	User data or supplier specific
STmin	20ms	0ms	0ms direct on CAN-Bus without Gateway and 5ms through Gateway	10ms

Table 3-23 STmin values specified by OEM

3.5.3 Network Layer Protocol

The network layer protocol shall use the frames as defined in Section Network Layer Frames for data transfer between network nodes. To initiate a data transmission with more than 6 bytes the sending node transmit a first frame. Before the sending node is allowed to send more data, it waits for a flow control frame from the receiving node. Based on the information in the flow control frame the sender transmit consecutive frames according to block size and minimum separation time as described in section Consecutive Frame and section Flow Control Frame. Figure Flow Control Mechanism shows a sequence diagram that illustrates a segmented transmission. After sending a block of consecutive frames the transmitter waits for another flow control frame from the receiving node. The last block of consecutive frames may contain less frames depending on the amount of data that is to be transmitted. After the last block of consecutive frames, the receiver does not send any flow control frames more.

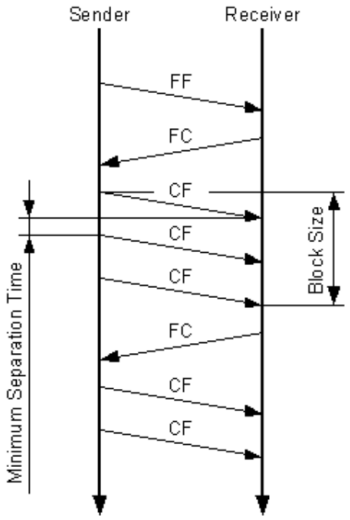


Figure 3-7 Flow Control Mechanism

3.5.4 TP CAN Identifier

The CAN-Identifiers for TP are the same as the CAN-Identifiers specified for diagnostic purpose. For details, refer to documents [QD1] and [QD2].

3.5.5 Data length Code (DLC)

The DLC parameter specifies the number of data bytes transmitted in a CAN frame according to [NS1]. The network layer in [NS6] does not specify any requirements concerning the length of the data field in a CAN frame other than those implied by the size of the network layer protocol data units. An application that implements the network layer as defined in [NS6] might either pad all CAN frames to their full length or optimize the DLC to the applicable length of the network layer PDU according to [NS6].

3.5.6 Flow control

In the following section, the timings and the timeouts for the TP are specified. In the illustration below the data flow using FC is reproduced.

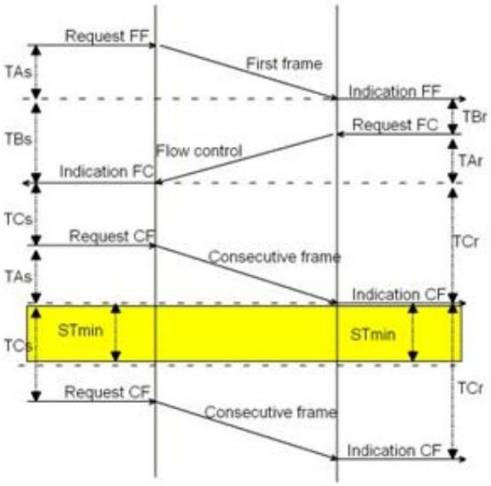


Figure 3-8 Data flow using timing parameters

The description of timing parameters listed in Table: Data flow using timing parameters can be found below.

TA_s - Time for transmission of the CAN frame on the sender side
TA_r - Time for transmission of the CAN frame on the receiver side
TB_r - Time until transmission of the next Flow Control (FC)
TB_s - Time until reception of the next Flow Control (FC)
TC_s - Time until transmission of the next Consecutive Frame (CF)
TC_r - Time until reception of the next Consecutive Frame (CF)
ST_{min} - The minimum time the sender is to wait between the transmissions of two CF

3.5.6.1 Network layer parameters

The table: Network layer timing parameter values [NS6] below specifies the network layer timing parameters to be used by default or if not otherwise specified for the external test equipment and ECUs. For legislated OBD/WWH-OBD communication refer to [NS4] and for vehicle manufacturer (OEM) enhanced diagnostic, the network layer timing parameter values as specified in [QD1] shall be used.

Parameter	Description	Timeout Value	Performance requirement value
N_As	Time for transmission of the CAN frame (any N_PDU) on the sender side	70ms	-
N_Ar	Time for transmission of the CAN frame (any N_PDU) on the receiver side	70ms	-
N_Bs	Time until reception of the next Flow Control N_PDU	150ms	-

N_Br	Time until transmission of the next Flow Control N_PDU	N/A	**(N_Br + N_Ar) < (0,9 × N_Bs)**
N_Cs	Time until transmission of the next Consecutive Frame N_PDU	<70ms	**(N_Cs + N_As) < (0,9 × N_Cr)**
N_Cr	Time until reception of the next Consecutive Frame N_PDU	150ms	-

Table 3-24 Network layer timing parameter values [NS6]

For more details on N_As, N_Ar, N_Br, N_Bs, N_Cs, N_Cr, refer to content of [NS6]. In some cases, "N" is substituted to "T", so N_As=T_As, N_Ar=T_Ar, N_Br=T_Br, N_Bs=T_Bs, N_Cs=T_Cs and N_Cr=T_Cr.

3.6 Transport, Session and Presentation Layer

Transport, session and presentation layers are not applicable in this context, because the required functionalities of these layers (e.g. segmentation, transformation, de- compression and de-encryption) are sourced out to the underlying network layer or are not used.

3.7 Application Layer

Application layer services are usually referred to as diagnostic services, general Network handling or Services for measurement and calibration for example. The timing parameters for some part of the application layer and the session layer are specified in NS7 (ISO 15765-3) for example.

3.7.1 Diagnostic Communication and Services

The diagnostic communication protocol defines diagnostic test services, message command structure/format, and data parameter identifiers used in the communication exchange between external test tool and the on-board control unit. The diagnostic protocol services are mapped to application-specific diagnostic functions by the high-level application.

3.8 Overview of CAN-Network

3.8.1 Introduction

This section specifies the Network Management for communication among electronic control units (ECU) on road vehicles and is to be referenced by every ECU specification that incorporates interconnection for ECU communication. Application layer services are usually referred for example to as diagnostic services, general Network handling or Services for measurement and calibration. In praxis, interconnected ECUs from different suppliers influence each other on the same bus, even if they implement CAN-communication according to ISO specifications. A faulty ECU influences other ECUs and the entire system as well. The result is that too many interdependent DTCs in different ECU fault memories aggravate the complexity of system fault analysis. Because this situation has to be solved in the same way for all ECUs, there is a need to standardize network handling as its own functionality and to separate it from ECU- functionality.
Moreover, this so standardized network handling functionality, also called "Network Management", shall fulfill these general requirements:

- Activation/deactivation of the CAN-hardware (controller, bus transceiver) and identification of network configuration during the startup procedure. Monitoring network configuration while bus system is running.
- Bus diagnosis and providing relevant bus-status information for the whole network.
- Bus error correction/recovery after a bus-off.
- Synchronized transfer of all network nodes to the bus sleep mode (energy-saving mode if no communication is required).

3.8.2 Wakeup/shutdown of ECUs with Ignition on/off

It is some time misunderstood that no network management is required for ECUs, which are activated with "ignition ON". This is true, if all ECUs on the same bus are powered over Ignition line and they can all be powered on at the same time with "ignition ON" and powered off at the same time again with "ignition OFF". It could be easy to survey this situation for two or three ECUs, which are activated/deactivated on the same communication bus with "ignition ON/OFF", but it is no more trivial to handle more ECUs in this way. It is not necessary, that all ECUs shall wakeup at the same time, but a coordinated transition to sleep mode is fundamental and required. Even if ECUs wake up with "ignition ON", they can take different long times on "ignition OFF" to reach their sleep or shutdown modes. Due to their different requirements on and before shutdown, some ECUs need more time to complete their activities like saving data in EEPROM or waiting for completion of slow mechanical and physical processes.

NM_CL15ONOFF_001:

For more than three ECUs with different shutdown-sequences and -times on the same communication bus, it is required to implement network management functionality as specified in this section. Therefore, NM is required for all ECUs connected to CAN-Bus, which are able to communicate without ignition signal (CL15 = OFF) and for ECUs connected to CAN capable of sustaining their power supply, if ignition is switch off (CL15 = OFF).

Note:

ECUs, which are immediately shut off by switching ignition state to "OFF" (CL15=OFF) don’t need to implement NM, because they can’t take part in CAN communication at any time while ignition state is off (CL15=OFF). According to NM_FUNCTIONAL_001, any ECUs, which don’t implement NM will be considered as "ready to sleep" ECUs. Such ECUs use only CL15-line as power supply and are not connected in any way to another power supply-line like CL30 or CL30s.

3.8.3 Network Management Strategies

For Network Management use AUTOSAR-NM of decentralized and direct network management existing in field and on the market, where every ECU transmits a special network management message, which is reserved for the network management.

AUTOSAR-NM

- Startup / Shutdown of Network
- Controlled transition of all ECU’s to bus-sleep mode
- Remote sleep indication
- Partial Networking (e.g. ISO11898 part 6)

- User data handling
- Bus Load Reduction
- Detection of present nodes

NM_GENERAL_001:

For the ease of reuse and future development, it is required, that Network management for OEMVehicles shall be compliant with AUTOSAR-Network Management.

NM_GENERAL_002:

OSEK-NM and AUTOSAR-NM use different mechanisms which are not compatible. It is therefore not allowed to combine or use both variants of network management on the same bus.

NM_GENERAL_003:

In Small Car Platform project, VinFast mention to use AutoSAR V4.3.1.

For special case shall be discussed and confirmed by VinFast.

3.8.4 AUTOSAR-Network Management

The required AUTOSAR Network Management functionality for CAN ECUs in OEM-vehicles is specified in these two documents [NS4] (AUTOSAR-Specification). The functionality can be summarized in the following two functional requirements:

NM_FUNCTIONAL_001

See SWS_CanNm_00087 in AUTOSAR - Specification of CAN Network Management [NS5]. Every network node in a CanNm cluster shall transmit periodic Network Management PDUs as long as it requires bus-communication; otherwise is shall transmit no Network Management PDUs.

NM_FUNCTIONAL_002

See SWS_CanNm_00088 in AUTOSAR - Specification of CAN Network Management (NS 5). If bus communication in a CanNm cluster is released and there are no Network Management PDUs on the bus for a configurable amount of time determined by CANNM_TIMEOUT_TIME + CANNM_WAIT_BUS_SLEEP_TIME (both configuration parameters) transition into the Bus-Sleep Mode shall be performed.

NM_FUNCTIONAL_003:

To fulfill the two requirements NM_FUNCTIONAL_001 and NM_FUNCTIONAL_002, it is required to use the appropriate functions as specified in SWS_CanNm_00032 in NS 4 (AUTOSAR - Specification) of Network Management Interface. At application level:

- Nm_NetworkRequest(…)
- Nm_NetworkRelease(…)

3.8.4.1 AUTOSAR-Network Management States and behavior

The main purpose of network management is to provide controlled wake-up and shutdown functionality for the entire communication network. The behavior of nodes participating in network management is

described as a state machine (see also Figure Network Management States). A node shall behave differently on the bus depending on its current network management state.
More details about states and transitions are provided in NS 4 and NS 5 (AUTOSAR - Specification).
The following figure shows an UML (Unified Modeling Language) state diagram with respect to the API specification. Mode change related transitions are denoted in green, error handling related transitions in red and optional node detection related transitions in blue. Additionally, it is assumed that busload reduction functionality is enabled.

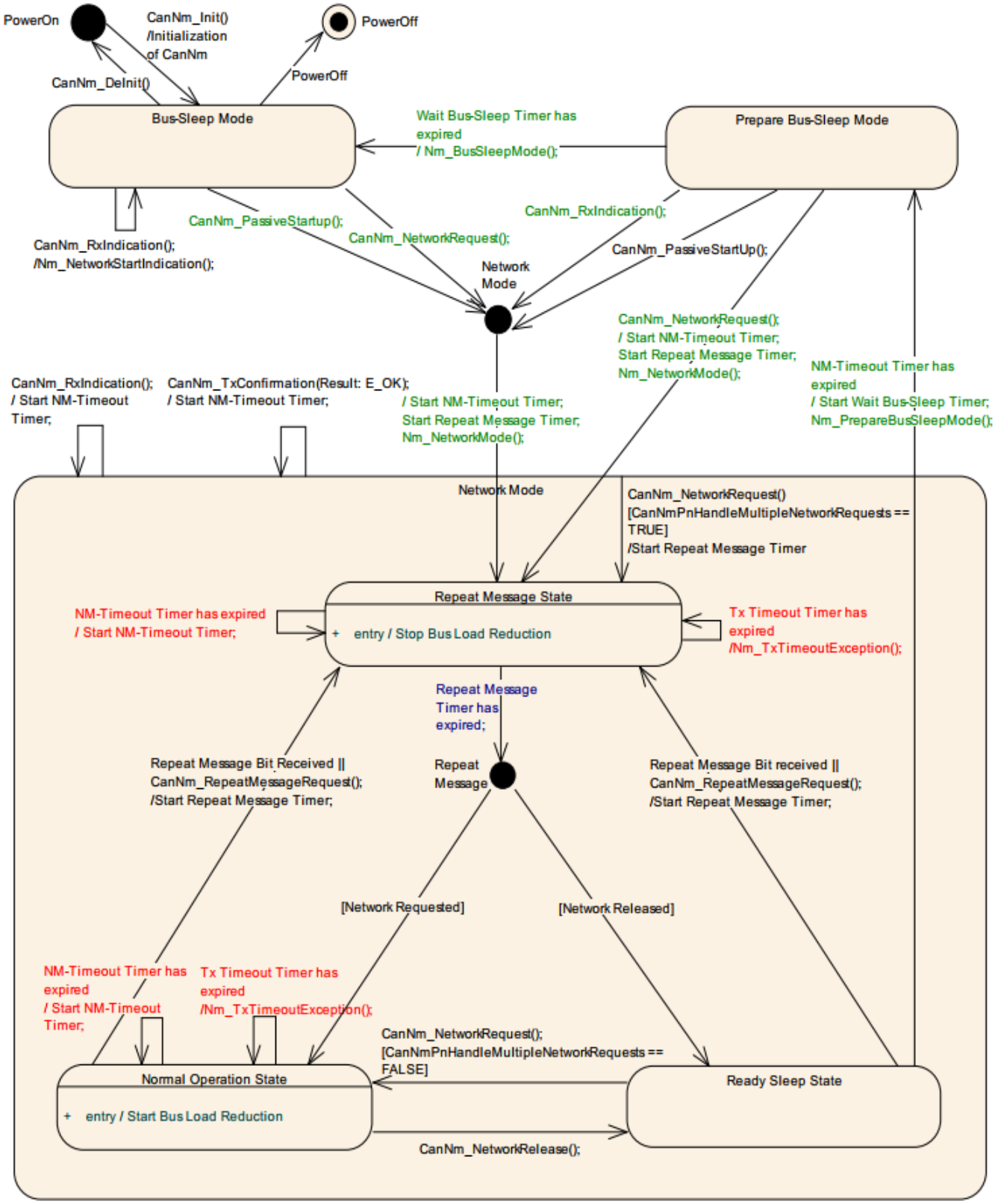


Figure 3-9 CAN Network Management Algorithm

Network management shall be initialized to *bus sleep* state. In this state the node shall not send any messages on the bus.

Upon an internal valid wake-up event, node shall request network by transmitting its NM PDU on bus, communication network management shall transit into *repeat messages* state.

When Node enters into the Repeat Message state, Node shall begin to transmit its NM PDU as fast as possible after wakeup and after First NM PDU trasnsmission ECU shall start sending normal application messages. Refer to **Figure 3-12** Detailed bus wake-up procedure.

Node shall NOT wakeup after receiving the application messages during sleep state.

When the Repeat Message State is entered from BusSleep Mode, Prepare Bus-Sleep Mode, Normal Operation or Ready Sleep State the CanNm module shall deactivate the busload reduction. (see content in SWS_CanNm_00052 of NS 5, AUTOSAR - Specification).

After the repeat message time-out (CANNM_REPEAT_MESSAGE_TIME) has elapsed, the node shall switch into normal or ready sleep states depending on network request state (i.e. requested or released). The network request state shall be set by the ECU itself depending on whether it requires bus communication or not.

When the Normal Operation State is entered from Repeat Message State or Ready Sleep State and CanNmBusLoadReductionEnabled is TRUE the CanNm module shall activate the busload reduction. The bus load reduction algorithm is described in detail in Section Bus Load Reduction Algorithm.

If the node is ready for shutdown it shall inform the network management by releasing the network.
Upon network release the network management transitions from normal state into ready sleep state. In this state the node shall monitor the bus for network management messages and if no network management messages are received for a certain amount of time (CANNM_TIMEOUT_TIME) it shall switch into prepare bus sleep state.

In prepare bus sleep state all nodes shall flush their transmission queues and stop transmission afterwards. After the bus sleep time-out CANNM_WAIT_BUS_SLEEP_TIME) has elapsed all nodes shall transit into bus sleep state at almost the same time (depending on node specific tolerances).

If a network management message with repeat message bit indication set is received, the network management shall transit into repeat messages state. Furthermore, a node's application software shall be able to request the repeat message state. In this case the network management shall transit into repeat messages state and transmit the next network management message with repeat message bit indication set.

For more details on AUTOSAR-NM- Behavior and implementation, see content of: NS 4 (AUTOSAR - Specification) and NS 5 (AUTOSAR - Specification).

Following table shows the bahaviour of the ECU for message transmision and Receiption in diffrent network modes

NM Mode		NM Frame/Message		App Frame / Messages	
		Tx	Rx	Tx	Rx
Bus Sleep Mode		N	Y	N	N
Prepare Bus Sleep Mode		N	Y	-	-
Network mode	Repeat Message Mode	Y	Y	Y	Y
	Normal Operation Mode	Y	Y	Y	Y
	Ready to sleep state	N	Y	Y	Y
"N" indicates that Tx/Rx of frame is NOT permitted "Y" indicates that Tx/Rx of frame is permitted. "-" denotes that frame of Tx/Rx shall not be prohibiting Note : App frame icludes application messages, diagnostic messages and calibration messages.					

Table 3-25 : ECU behavior for Msg Rx & Tx

3.8.4.1.1 Timing Parameter for AUTOSAR-NM

NM_TIMING_001:

In Table Network management timing parameters, the values of the timing parameters mentioned in the previous sections are listed.

Parameter	Value
CANNM_TIMEOUT_TIME	2000 ms
CANNM_WAIT_BUS_SLEEP_TIME	1500 ms
CANNM_REPEAT_MESSAGE_TIME	3200 ms
CANNM_MSG_CYCLE_TIME	640 ms
CANNM_MSG_CYCLE_OFFSET	0 ms – 635 ms
CANNM_MSG_REDUCED_TIME	320 ms – 637,5 ms

Table 3-26 Network management timing parameters

NM_TIMING_002:

The node specific parameters CANNM_MSG_CYCLE_OFFSET and CANNM_MSG_REDUCED_TIME shall depend on the Node identifier (NODE_ID) assigned for each node and the values shall be computed using the following rules:

$$CANNM_MSG_CYCLE_OFFSET = NODE_ID * 5\,ms$$

$$CANNM_MSG_REDUCED_TIME = 320\,ms + NODE_ID * 2.5\,ms$$

3.8.4.1.2 NM configuration Parameter

NM_CFG_001:

In Table 3-27: NM configuration switches the values of the configuration parameters are listed and explained.

Configuration switch	Value	description
NM_USER_DATA_ENABLED CANNM_USER_DATA_ENABLED	ON	Enable User Data as required for bus diagnostic and partial networking Note : Partial networking is NOT planned for VINFAST program unless its specified in specifications
NM_BUS_SYNCHRONIZATION_ENABLED CANNM_BUS_SYNCHRONIZA TION_ENABLED	ON	only for ECUs with Gateway Functionality. If an ECU doesn't implement a Gateway-functionality(e.g. NM-Coordinator) this value shall be set to OFF
NM_REMOTE_SLEEP_IND_ENABLED	OFF	Except ECUs for which, it is explicitly required to enable "remote sleep indication".
NM_NODE_DETECTION_ENABLED, CANNM_NODE_DETECTION_ENABLED	ON	Enable by default to be able to further use the functionality without modification of NM-Configuration
NM_CONTROL_BIT_VECTOR_ENABLED CANNM_CONTROL_BIT_V ECTOR_ENABLED	ON	Extension for partial networking. This is ON to extract repeat message request bit from Control bit vector Note : Partial networking is NOT planned for VINFAST program unless its specified in specifications
NM_BUS_LOAD_REDUCTION_ENABLED	OFF	Disabled According to NM_BUSLOA D_001. Each ECU send its NM-PDU cyclically if it required network communication
CANNM_IMMEDIATE_RESTART	ON	To enable immediate transmission of an NM message if an ECU requires the network in "Prepare Bus Sleep"

Table 3-27 NM configuration switches

3.8.4.1.3 AUTOSAR-Network Management Messages

NM_MSG_001:

For network management communication, every node (ECU) shall be assigned a unique identifier (NODE_ID). For simplicity it is recommended to use the same identifier for network management as used for diagnosis. The node identifier (NODE_ID) shall be an eight bit value of the range 0x00 (0 dec)

to 0x7F (127 dec). The node identifier together with the 3 bit class identifier (as defined in section CAN communication for network management) shall be used to determine the CAN identifier for network management message of each NM-Node (ECU).

0x500 < (CAN NM-Message Identifier = 0x500 + NODE_ID) <= 0x57F

NM_MSG_002:

The NM-PDU (Message) shall have 8 bytes (DLC=8) according to the format in Figure Network management message format, in which the unique node identifier(NODE_ID) shall be transmitted in the signal "srcNodeId (byte 0)".

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 7	User Data 5 (usrData5)							
Byte 6	User Data 4 (usrData4)							
Byte 5	User Data 3 (usrData3)							
Byte 4	User Data 2 (usrData2)							
Byte 3	User Data 1 (usrData1)							
Byte 2	User Data 0 (usrData0)							
Byte 1	Control Bit Vector (CtrlBitVector)							
Byte 0	Source Node Identifier (srcNodeId)							

Figure 3-10 Network management message format

NM_MSG _003:

The repeat message request (RMR = RptMsgRequest) bit 0 of the control bit Vector (CtrlBitVector = byte 1) is used, if set to 1 to force the network management of all other nodes on the bus into the repeat message state.

At Repeat Message Request in the Normal Operation State, the NM module shall enter the Repeat Message State, the NM module shall set the Repeat Message Bit.

At Repeat Message Request in the Ready Sleep State, the NM module shall enter the Repeat Message State. the NM module shall set the Repeat Message Bit.

Repeat Message Request will be triggered when it still need to communicate because of node detection (eg. GW synchronous bus sleep).

When Repeat Message State is left, the NM module shall clear the Repeat Message Bit.

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CBV	reserved	Partial Network Information Bit	reserved	Active Wakeup Bit	NM Coordinator Sleep Ready Bit	Reserved	Reserved	Repeat Message Request

Figure 3-11 Control Bit Vector in the network management message format

The Control Bit Vector shall consist of
Bit 0: Repeat Message Request
0: Repeat Message State not requested
1: Repeat Message State requested
Bit 3: NM Coordinator Sleep Bit
0: Start of synchronized shutdown is not requested by main coordinator
1: Start of synchronized shutdown is requested by main coordinator
Bit 4 Active Wakeup Bit
0: Node has not woken up the network (passive wakeup)
1: Node has woken up the network (active Wakeup)
Bit 6 Partial Network Information Bit (PNI)
0: NM PDU contains no Partial Network request information
1: NM PDU contains Partial Network request information
Bit 1, 2, 5, 7 are reserved for future extensions
0: Disabled / Reserved for future usage.

NM_MSG _004:

The six user data bytes (byte 2,3,…,7) are reserved for OEM custom extensions of network management. Transmitting nodes shall always send network management messages with eight bytes where these six data bytes are set according to definition in section NM-User Data.

3.8.4.1.4 Bus Wake-up

NM_WAKEUP_001:

Network nodes woken-up via external events that require bus communication shall transmit a network management message to signalize its need to communicate on the Bus. A CAN transceiver with wake-up functionality that observes the wake-up pattern on the bus shall activate the ECU- application, if all activation conditions for ECU are fulfilled. The ECU-application shall transmit a network management message in this case, if it needs to communicate on the Bus.

NM_WAKEUP_002:

Due to differing implementations of CAN transceivers, a CAN transceiver may also activate the ECU on shorter dominant states observed on the bus. But to prevent ECUs from unintentional wake-up, the dominant time for wake-up via bus of the used transceiver is defined to be between 0.75µs and 5µs.

NM_WAKEUP_003:

On startup each ECU shall send its NM-message not later than the availability time as defined in the Wakeup-timing-parameters.
Activate Immediate NM Burst mechanism if ECU is awakened from the Bus sleep state due to network request.
If the Repeat Message State is not entered via Network Request OR Immediate Nm Burst, the transmission of NM PDU shall be delayed by CANNM_MSG_CYCLE_OFFSET after entering the repeat message state.

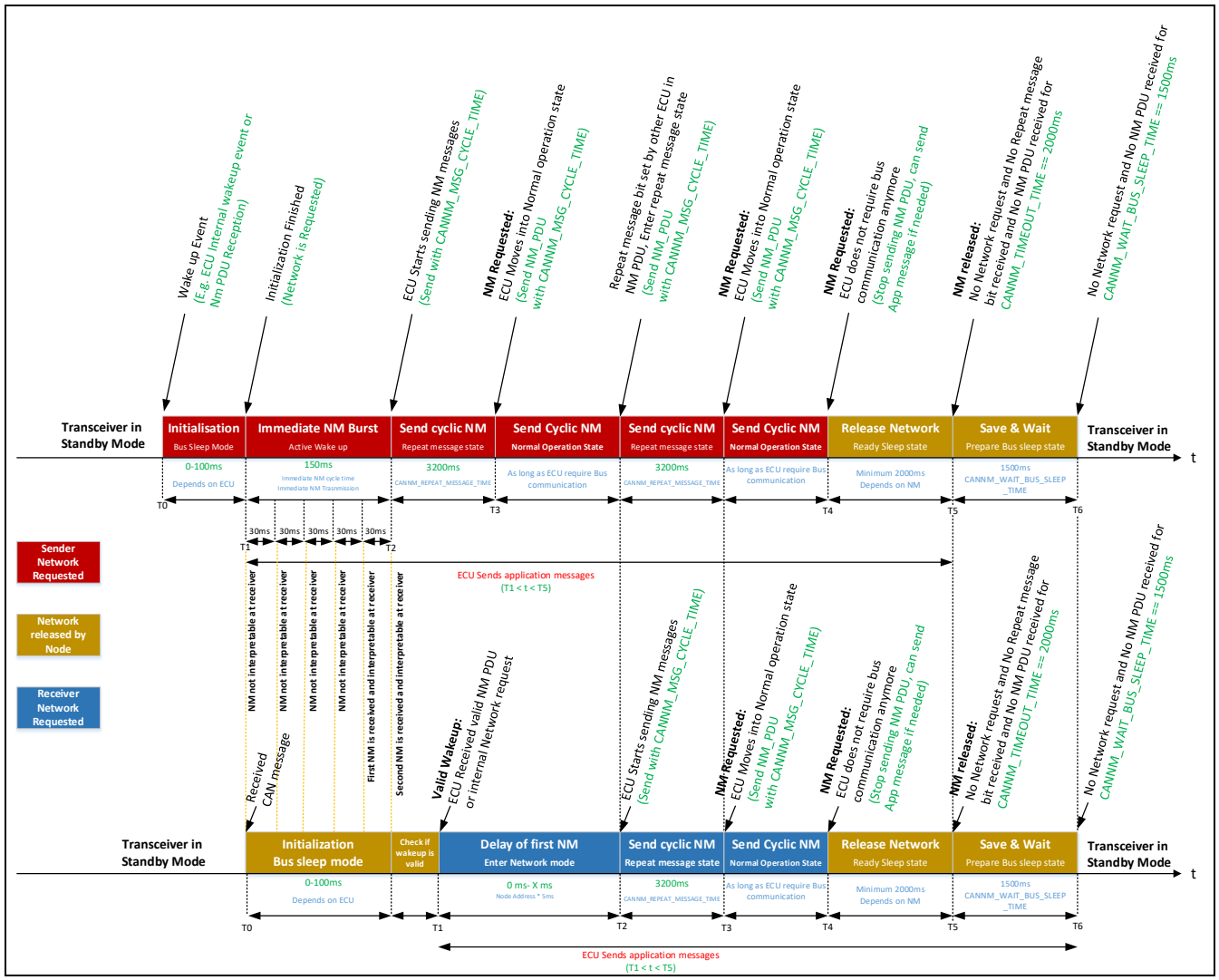


Figure 3-12 Detailed bus wake-up procedure

T0	CanNM module receives a wake-up event and starts initialization.
T1	CanNM module is initialized and can interprete signals on bus after t > T1.
T2	If wake up is valid and ECU specific delay time (ECU ID*5ms) has run out, CanNM module enters Network state – Repeat message state.
T3	Leave repeat message state: Network requested: Go to Normal operation state Network released: Go to Ready sleep state
T4	Network is released: Go to ready sleep state. OR Network has never been requested (timeout). Stay in bus sleep mode.
T5	No network requested AND No NM received in Ready sleep state for CANNM_TIMEOUT_TIME. Go to prepare bus-sleep mode.

	OR Network has never been requested (timeout).
T6	Network requested: Go to Repeat Message state OR Network (still) released: No Network request (ECU internal) AND No valid NM received for CANNM_WA IT_BUS_SLEEP_TIME Go to Bus-sleep mode.

Table 3-28 Wake-up timing parameters

3.8.4.1.5 Bus Load Reduction Algorithm

The main purpose of bus load reduction- algorithm is to reduce the bus load to a defined level that is independent of the number of nodes on the network. During bus load reduction the transmission of network management messages shall be controlled by received network management messages. If a network management message is transmitted by a node it resets its message transmission time-out to the global network management cycle time (CANNM_MSG_CYCLE_TIME). Upon reception of a network management message the receiving nodes resets their message transmission time-out to a node specific value shorter than the global network management cycle time (CANNM_MSG_REDUCED_TIME). A node only transmits a network management message if its transmission time-out has elapsed. Due to the node specific reduced message cycle times this algorithm guarantees that at most two network management message are transmitted within the global network management cycle time. On the other hand it allows every node to keep the network alive as long as it requires bus communication.

NM_BUSLOAD_001:

The bus load reduction algorithm shall only be activated in the normal operation state, if required to be enabled in the future. In the current specification, bus load reduction shall be disabled.

3.8.5 NM-User Data

In NM-PDU, there are six (6) bytes called "User Data" reserved for user or OEM specific usage. They are supplementary ECU-application specific piece of data that is attached to every NM message sent on the bus. The usage of the user data is not standardized, so their usage can be a limitation for future extensions or modifications. User data are very helpful for analysis of global fault of ECU startup or sleeping instances and therefore, they shall be documented in the communication matrix.

NM_USRDATA_001:

User data support shall be enabled for each network node (ECU) that implements network management.

NM_ USRDATA _002:

User data shall be read and written at application level using the functions specified in SWS_CanNm_00035, SWS_CanNm_00132 in NS 4.

- Nm_GetUserData(…)
- Nm_SetUserData(…)

NM_USRDATA_003:

By default, all six bytes of user data shall be set to 0xFF (255 dec) indicating, that they are not used for a specific purpose, unless there is another definition.

NM_USRDATA_004:

The first user data byte (usrData0 = byte 2 of NM-PDU) indicates the implemented functionality of user data. This value is to be acted as a version number, that tell how to interpret the content of the remaining five user data bytes (usrData1, usrData2, …, usrData5). This value is always constant for given software version.

usrData0	Description/Interpretation
0xFF	User data not used
0x00	User data for current vehicle without partial networking
0x01	User data for current vehicle with partial networking

Table 3-29 Interpretation of usrData0 as functionality of user data

NM_USRDATA_005:

Wakeup reason and the reason to be awake shall be provided in usrData1 from the application of every ECU supporting network management at any time when the network is awake. The ECU transmits the application reason that cause the CAN network to wakeup. When CAN communication causes an ECU to wake up, the wakeup reason shall be "Wakeup on Network".
ECUs shall set this wakeup reason "NETWORK_WAKEUP" on ECU- startup. After Startup delay time the wakeup reason "NETWORK_WAKEUP" shall be cleared. The wake up reasons in the following table are predefined for every NM-node if usrData0 = 0x00 or usrData0 = 0x01. ECU specific wake up reason can be signaled by using ECUSPEC_WAKEUP.

usrData1	Definition	Description/Interpretation
Bit 0	IGNITION_WAKEUP 0x0 inactive 0x1 active	ECU was "power off" or "Bus sleep" mode and started by Ignition turn on
Bit 1	RESET_WAKEUP 0x0 inactive 0x1 active	If ECU was already awake, in "Prepare Bus Sleep" or "Ready Sleep" mode and re-started by an internal or an external reset.
Bit 2	NETWORK_WAKEUP 0x0 inactive 0x1 active	ECU was in "Power off", "Bus Sleep", "Prepare Bus Sleep" or "Ready Sleep" Mode and then re-started by a network communication.
Bit 3	ECUSPEC_WAKEUP 0x0 inactive 0x1 active	ECU was in "power off", "Bus Sleep", "Prepare Bus Sleep" or "Ready Sleep" mode or "awake" and re- started by ECU-specific reason (e.g. pressed Button, timer, etc.)

Table 3-30 Defined wake up reasons

ECUs shall transmit all current application reasons which prevent the ECU from releasing the network. It shall transmit the reasons for keeping the bus awake permanently in the srData1.as long as the ECU has not released the network. The Stay Awake reasons in the following table are predefined for every NM-node if **usrData0 = 0x00** or **usrData0 = 0x01** ECU specific reason can be signaled by using ECUSPEC_AWAKE.

usrData1	Definition	Description/Interpretation
Bit 4	NETWORK_AWAKE 0x0 inactive 0x1 active	ECU need network communication and keeps the network awake
Bit 5	IGNITION_AWAKE 0x0 inactive 0x1 active	ECU can't release communication because Ignition is turn on
Bit 6	DIAGNOSTIC_AWAKE 0x0 inactive 0x1 active	ECU need communication for diagnostic purpose and keeps network awake.
Bit 7	ECUSPEC_AWAKE 0x0 inactive 0x1 active	ECU keeps the network awake for ECU-define reasons for which communication is needed

Table 3-31 Defined awake reasons

3.8.6 NM at Gateways

A gateway holds for every bus with a network management channel and the coordination does take place over a NM master function. During the start of the bus communication there are no special requirements necessary. It is necessary to wake up all buses on the gateway synchronously. For communication shutdown on the buses, which are connected over gateway with NM, an enlargement of the NM is necessary. Especially the requirement to shut down all buses synchronously makes it necessary to enlarge the NM. It is necessary to mirror the shutdown readiness of one bus to the other bus which is connected by the gateway to another bus. In AUTOSAR-NM, a so called NM-Coordinator functionality is introduced to prepare the information whether one or more nodes are not ready to sleep or every nodes on all connected buses are ready to sleep. For Details on Requirements of Network Management on Gateway-ECU see section CAN -wakeup.

3.8.7 NM and Diagnostic communication control services

As required in NM_GENERAL_001, ECU shall transmit periodic NM PDUs (Messages) as long as it requires bus-communication, but in some situation, it is required for programming purpose for example, to disable normal communication including NM-Messages transmission during programming session. After Programming session, it is necessary to enable again the normal communication and NM-messages transmission.

NM_DIAG_001:

To disable and re-enable NM-messages in diagnostic sessions, the AUTOSAR-NM provides two API-Functions that shall be used for this purpose. See SWS_CanNm_00033 in NS 4.

- Nm_DisableCommunication(…)
- Nm_EnableCommunication(…)

3.9 CAN Wakeup

3.9.1 Introduction

The purpose of this section is to define requirements of the timing of CAN messages on the CAN bus system for road vehicles

3.9.2 Network Wakeup Strategy

This following sub-sections specifies strategies of the wake-up process of ECUs.

3.9.2.1 Timing Specification undesired wakeup

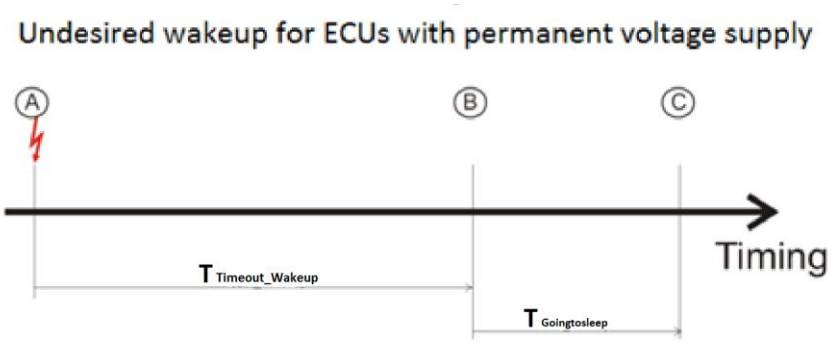


Figure 3-13 Undesired wakeup for ECUs with permanent power supply

The following Table: Descriptions of time-intervals for undesired wake-up of ECUs with permanent power-supply describes the meaning of the different points of the time-line.

Ⓐ	Trigger at bus physics or detection of any wakeup event (e.g. button pressed, timer, etc.)
Ⓑ	At this time point, ECU shall have finished to check the of validity of the wakeup event. The Network modes are specified in documents [NS4] and [NS5].
Ⓒ	If no NM-message was received and ECU don't need network communication itself, it shall leave network mode and go back to sleep mode.

Table 3-32 Descriptions of time-points for undesired wake-up of ECUs with permanent power-supply

$T_{Timeout_Wakeup}$	In this time interval, the ECU has to check the validity of wakeup events to enter its network mode section Overview of CAN-Network as well as specifications: [NS4] and [NS5]. If there is no event valid for Bus wakeup, the ECU is not allowed to enter its network mode.
$T_{Goingtosleep}$	In this time interval, ECU can stay in network mode if ECU don't need network communication itself and no NM-message was received, it has to leave

	network mode and go back to bus sleep mode. If a NM-message was received and the ECU was not in network mode, it has to re-enter network mode.

Table 3-33 Descriptions of time-intervals for undesired wake-up of ECUs with permanent power-supply

Note:
Based on this, a "Startup delay time" is defined as the time frame within the application can release the network, if the communication is no more required.

TStartupDelayTime = TTimeout_Wakeup + TGoingtosleep

Name	Min. Time [ms]	Typ. Time [ms]	Max. Time [ms]
$T_{Timeout_Wakeup}$	3000	CANNM_REPEAT_MESSAGE_TIME = 3200	4000
$T_{Goingtosleep}$	NA	CANNM_TIMEOUT_TIME + CANNM_WAIT_BUS_SLEEP_TIME = 3500	4000

Table 3-34 Timing table undesired wakeup

Deviations shall be discussed and agreed by OEM.

3.9.2.2 Readiness for communication or Bus sleep

To ensure the operational readiness of the overall system (whole vehicle) in a reasonable time, it is required for each ECU in the vehicle communication network to have a maximum start-up time as specified.

3.9.2.3 Wake up

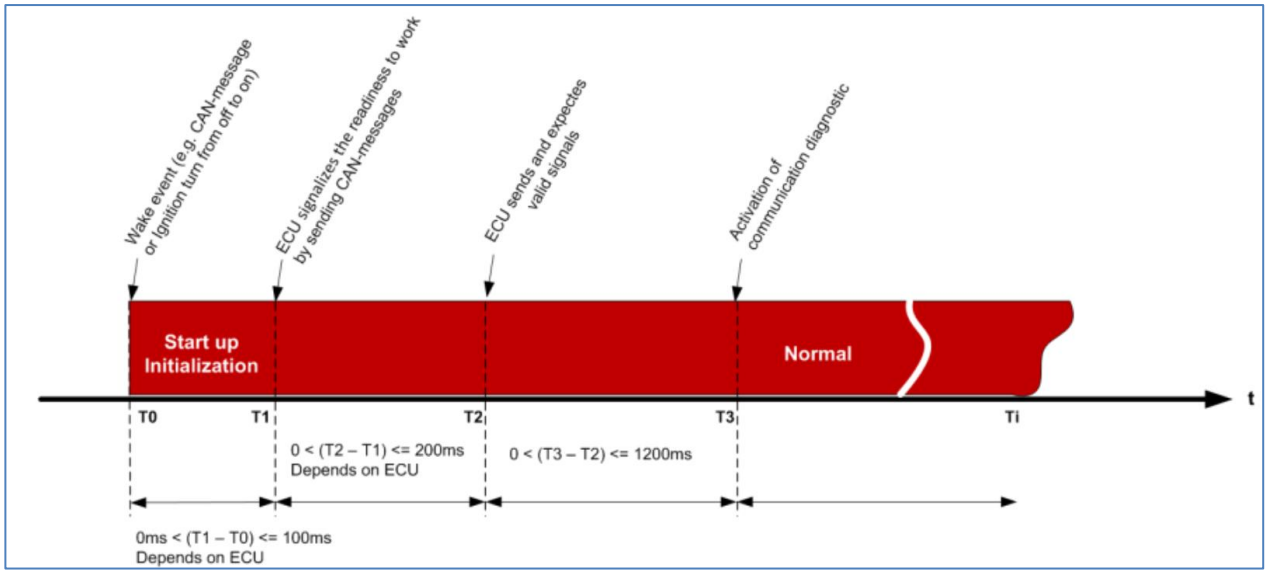


Figure 3-14 From Wake up until activation of communication diagnostic

3.9.2.4 Bus sleep

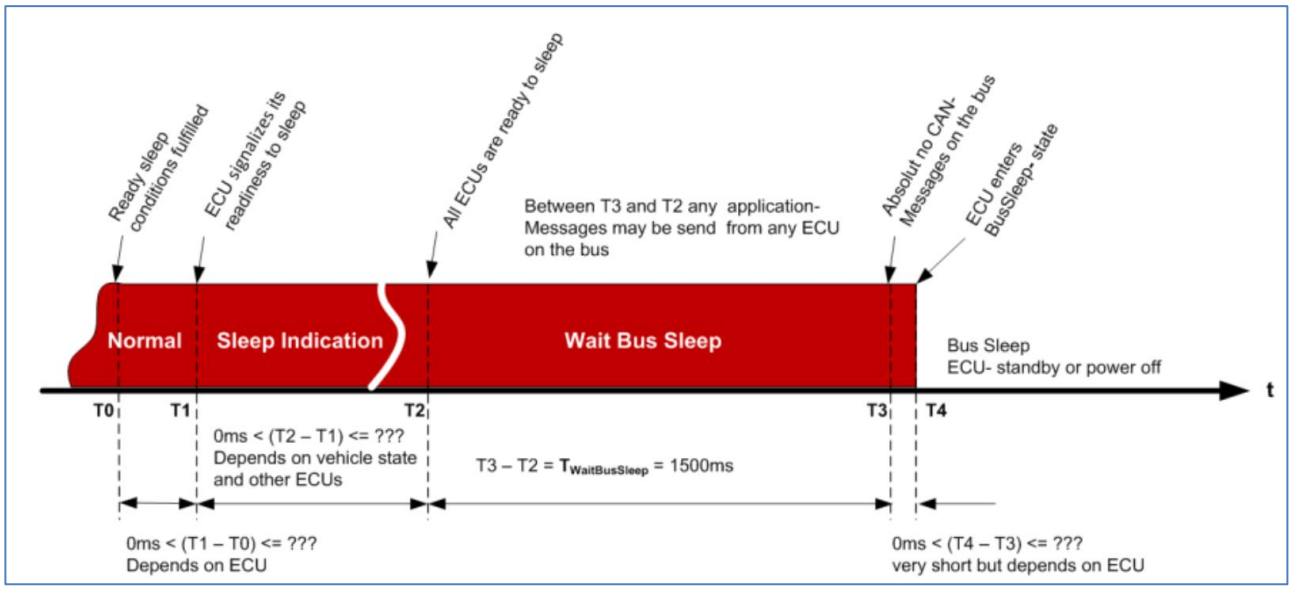


Figure 3-15 From "Ready sleep" until "Bus sleep"

3.10 Safety and availability requirements

COM_SAFEAVAIL_001

Every network node (ECU) shall ensure/guarantee proper communication for power supply voltage range defined in [QD10]. section Operation voltage ranges

COM_SAFEAVAIL_002

Any ECU- internal error/fault shall not affect the bus communication (e.g. sending error frames or put the bus permanently in a "dominant state"). For details see [QD4]

COM_SAFEAVAIL_003

A high bus load on the bus strand shall not cause disturbances or perturbations on ECU functionality or put the ECU in an unsafe state.

COM_SAFEAVAIL_004

Error frames on a bus strand shall not influence the node (ECU) internal functionality or put it in a unsafe state.

3.10.1 Acceptance of RX-messages

Each ECU shall only receive and monitor its received(RX)- signals as specified in the communication matrix. This mean, that no message shall pass the node HW-message filter, if its content is not needed in the ECU-application.
In case of there are too much Message transceiver, consider to use SW-message filter after get the confirmation from VinFast.

3.10.2 Monitoring system voltage

Each ECU shall receive and monitor the system voltage value via the appropriate bus communication message, even if system voltage can be directly measured.

NOTE:

An exception applies to the ECU that transmits the system voltage on the bus.

3.10.3 Monitor odometer-value

Each ECU shall receive and monitor for the vehicle odometer value via the appropriate bus communication message.

NOTE:

An exception applies to the ECU that is transmitting the odometer value on the bus.

3.10.4 Wrong odometer-value

If the bus communication has failed or an invalid odometer value is received, the ECU shall use the last known valid odometer value for the current power cycle, see [QD4].

3.10.5 Monitoring state of ignition

Each ECU shall receive and monitor the ignition status value (CL15) via the appropriate bus communication message, even if ignition status can be directly measured (e.g. hardwired ignition feed).

NOTE:

An exception applies to the ECU that has the ignition switch hardwired to it and is transmitting the ignition status on the bus.

3.11 CAN-Communication

Bus inputs are defined as signals in messages received by an ECU from the bus and shall be interpreted as specified in the communication matrix. The following requirements apply to ECU bussed inputs.

3.11.1 Signal Specification

A signal is a bus input and indicates a single part of CAN-message, defined by a resolution in bits and a formula to calculate the value. For representation of signal values, different data types as specified bellow are used.

3.11.1.1.1 Bit Order

According to the CAN bus protocol, the bit order is a **big endian mode**, that is, **msb first**, so there is no need to distinguish.

In a digital data bit string, the Msb is a bit of the highest digit, and the Lsb is a bit of the lowest digit. Digital data is binary, and like ordinary numerical notation, the left end is the highest digit, while the right end is the lowest digit. For example, 99 in the decimal system is expressed as (Msb)01100011(Lsb) in the binary system. In this case, the Msb is 0 and the Lsb is 1. Considering inter-block interfaces, with a parallel interface, all the bits on the transmitting side and the receiving side correspond, so there is no problem in particular. However, with a serial interface, all the bits are transmitted and received through a single data line, so that the bit order definitions in the protocols of the transmitting side and the receiving side must match. In the case of the above data (number), the data is transmitted and received in the order of "01100011" in the case of Msb first:

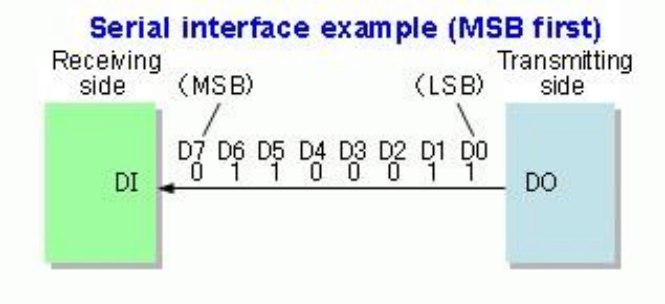


Figure 3-16 example for most significant bit

3.11.1.1.2 Byte Order

COM_SIGMSBMSB_001:

All signals shall be represented in **Motorola forward MSB** format (big endian). That means the most significant byte (MSB) shall be transmitted at first and the least significant byte (LSB) shall be transmitted at last.

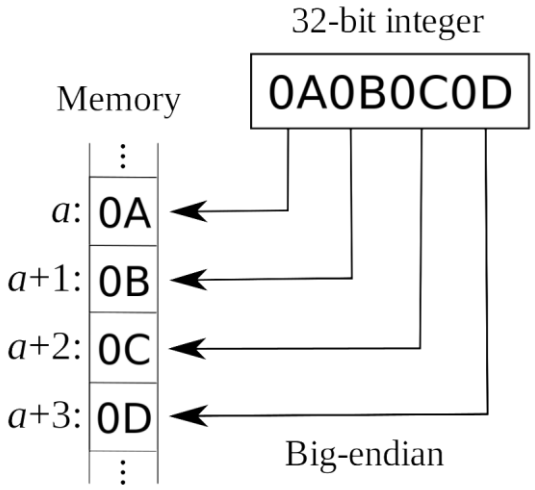


Figure 3-17 Example for Motorola byte order

3.11.1.1.3 Data Types

On CAN different data types with varying size may be used.

3.11.1.1.3.1 Unsigned Integer

COM_SIGDATATYPE_001:

Data type for representation of positive integer values like counting values for example, which always begin with zero (0).

Phy = Hex

Where

$0 <= Phy <= 2^b - 1$

b = 2,3, …, 64 = number of bits representing the unsigned integer value.

Number of bits	Min		Max	
	phy	Hex	Hex	Phy
2	0	0x0	0x3	3
4	0	0x0	0xF	15
8	0	0x00	0xFF	255
12	0	0x000	0xFFF	4095
16	0	0x0000	0xFFFF	65535
24	0	0x000000	0xFFFFFF	+16 777 215
32	0	0x00000000	0xFFFFFFFF	+4 294 967 295
48	0	0x000000000000	0xFFFFFFFFFFFF	+281 474 976 710 655
64	0	0x0000000000000000	0xFFFFFFFFFFFFFFFF	+18 446 744 073 709 551615

Table 3-35 Example Range of Unsigned Integer Values

3.11.1.1.3.2 Signed Integer

COM_SIGDATATYPE_002:

"Signed integer" as specified there shall be used to represent signal, where the signal value range includes negative values. To be able to transmit negative values, unsigned integer with an offset shall be used according to following formula.

Phy = Hex - $2^{(b-1)}$

Where

b = 2,3,… 64 = number of bits representing the "signed integer" value

0 <= Hex <= **2^b - 1**

$-(2^{(b-1)}) <= Phy < (2^{(b-1)} - 1)$

Number of bits	Min		Max	
	phy	Hex	Hex	Phy
2	-2	0x0	0x3	+1
4	-8	0x0	0xF	+7
8	-128	0x00	0xFF	+127
12	-2 048	0x000	0xFFF	+2 047
16	-32 768	0x0000	0xFFFF	+32 767
24	-8 388 608	0x000000	0xFFFFFF	+8 388 607
32	-2 147 483 648	0x00000000	0xFFFFFFFF	+2 147 483 647
48	-140 737 488 355 328	0x000000000000	0xFFFFFFFFFFFF	+140 737 488 355 327
64	-9 223 372 036 854 775 808	0x0000000000000000	0xFFFFFFFFFFFFFFFF	+9 223 372 036 854 775 807

Table 3-36 Example Range of Signed Integer Values

3.11.1.1.4 ASCII-Code

COM_SIGDATATYPE_003:

Data type for single characters representation in ASCII-Code. The ASCII-Code according NS 8 (ISO/IEC 8859-1) shall be used. This is the 8 bit single byte coded character set.

3.11.1.1.5 Enum

COM_SIGDATATYPE_004:

Enumerations are unique data types to define a list of different elements (e.g. Monday, Tuesday, Wednesday,..)

Symbol	Description	Coding raw data
N	Neutral	0x7 = 0111b
P	Parking	0x8 = 1000b
5	5th gear	0xE = 1110b
4	4th gear	0xC = 1100b
3	3th gear	0xD = 1101b
2	2nd gear	0x4 = 0100b
1	1st gear	0x2 = 0010b
	Invalid signal	0xF = 1111b

Table 3-37 Example of Enumeration

3.11.1.1.6 Fixed- and Floating-Point

COM_SIGDATATYPE_005:

Because the most ECUs do not have floating point processing units, the floating-point representation shall not be used for representation of Signal values in the CAN-communication in OEM-vehicle. Fractional values usually in base 10 are scaled and used as unsigned integer. For computing inside of the ECU, fixed-point representation is usually used.

3.11.1.2 Signal Scaling

COM_SIGSCALING_001:

A signal as define in the communication matrix shall have following properties, which are used to specify signals more precisely:

Signal property	Property Description
Short name or symbolic name	Symbolic name of signal. ASCII string with limited length usually used in communication matrix, ECU-application or measurement system as unique identifier of signal.
Description	This is comment or text to describe the signal by giving more information on how to use signal.
Size in bits	Number of bits use to represent the signal.
unit	This give the unit of physical variables, that the signal represent
Minimal value	Minimal possible value of the signal
Maximal value	Maximal possible value of the signal
Initial/Default value	Default or initial value of the signal. This usually needed on ECU startup
Invalid value	Unusable Value of signal. This is usually a constant value defined for the signal. This value will be sent for example, if no stable or valid value is available for the signal

Scale Factor	Factor use to scale the signal, so that it can fit in its size with limited number of bits
Scale offset	Like the scale factor, this offset is also used to scale the signal, so that it can fit in its size with limited number of bits

Table 3-38 Example of Signal Properties

Every signal specified in the communication matrix shall be or can be expressed with following scaling formula:

Phy = O + F * Hex

Hex = (Phy - O) / F

Where

Hex = raw value of signal send on communication bus

Phy = physical value signal used in specification documentation

F = Scale factor as described in Table: Example of signal properties. this shall not be 0

O = Offset as described in Table: Example of signal properties

Taking example of a signal X as ambient air temperature in vehicle, which is specified in the communication matrix as following:

symbolic name (short name) = STAT_AmbientAirTemperature

(min value = -40°C) <= **X** <= (max value= 85°C)

Size = 8 bits

F = 0.5

O = -40

[(-40 – (-40))/0.5] <= (**X** = STAT_AmbientAirTemperature) <= [(85 – (-40))/0.5]

(0 =0x00) <= (**X** = STAT_AmbientAirTemperature) <= (250 = 0xFA)

NOTE:

As expressed with specified value range and bit size, they are possible but not valid values of X in the range from (0xFB=85,5°) to (0xFF=87,5°C). ECU'-application shall take care of these invalid values if they have been received from CAN-bus or computed inside of the ECU.

Short name	MileAge		STAT_EngineSpeed			STAT_AmbientAirTemperature
Description	Actual kilometer reading		Engine speed			Actual ambient air temperature
Size in bits	24		16			8
unit	km		rpm			°C
Minimal value	0	0x000000	0	0x0000	-40	0x00
Maximal value	16777215	0xFFFFFF	16 383	0xFFFC	85	0xFA
Initial/Default value	0	0x000000	0	0x0000	-40	0x00

Invalid value	16777215	0xFFFFFF	**>** 16383	**>** 0xFFFC	**>** 85	**>** 0xFA
Scale Factor	1		0.25			0.5
Scale offset	0		0			-40

Table 3-39 Example of Signal Scaling

3.11.1.3 Signal out of Range or Signal not Valid

Invalid value also called "Signal Not Valid = SNV"-Value of a signal are values, which exceed the defined range of the signal. In most case this correspond to the maximum value, which can be represented with this signal.

COM_SIGSNV_001:

"Signal not Valid" value of a signal shall be expressed with the following general formula: **SNV = 2^b – 1** Where **b** is the number of bits needed to represent the signal (signal size).

Signal size [bits]	SNV-Value [dec]	SNV-Value [Hex]
2	3	0x3
3	7	0x7
4	15	0xF
6	63	0x3F
7	127	0x7F
8	255	0xFF
10	1 023	0x3FF
12	4 095	0xFFF
15	32 767	0x7FFF
16	65 535	0xFFFF
24	16 777 215	0xFFFFFF
32	4 294 967 295	0xFFFFFFFF
48	281 474 976 710 655	0xFFFFFFFFFFFF
64	18 446 744 073 709 551 615	0xFFFFFFFFFFFFFFFF

Table 3-40 Example of SNV-Values

3.11.1.4 Initial or Default Signal

COM_SIGSIVSDV_001:

In some cases, shortly after startup of an ECU, all signals of a CAN message may not be available and initial or default values shall be sent. The "Signal Initial Value = SIV" or "Signal Default Value = SDV" may be the same as the "Signal Not valid"- value, if not explicitly defined in vehicle communication matrix.

COM_SIGSIVSDV_002:

The "Signal Initial Value= SIV"-value shall be used at network startup by a transmitting ECU when a

signal is transmitted before a valid value has been measured or computed.

3.11.1.5 Signal not available / unused or empty bit

The vehicle communication matrix shall define a "Signal Not Available" -SNA- value for example for some signals computed or routed from Gateway. If an SNA is defined for a given signal, the application software should set its transmitted value to SNA when the input is un-known or becomes unavailable. Some situations where SNA may be used are when a sensor fails or a subsidiary bus, such as LIN, fails leaving the actual value of the signal unknown to the device that transmits the information on the CAN-communication bus. The term of "Signal Not Available" is also applicable for parts of a message, which are not used for a specific purpose (also called unused or empty bits in a message).

COM_SIGSNA_001:

unless otherwise defined, "Signal Not Available = SNA" value is defined as following:

SNA-Value = SNV–Value

3.11.1.6 Message Classes

A Message is the effective part of a Frame, that can be used by the application. They are raw bytes with max length and identifier as defined in the protocol specification. In addition to general specification, they are OEM specific message properties specified in the communication like message classes, categories, etc.

The CAN-messages are classified according to the following criteria:

- Dynamics of the messages
- Cyclic/Non cyclic messages
- Event triggered messages
- Reaction time on the bus

3.11.1.6.1 Class A

Fast messages, cycle time on HS-CAN

HS-CAN
t <= 30ms

Table 3-41 Class A

3.11.1.6.2 Class B

Messages with a medium cycle for HS-CAN, time critical, event triggered messages (tbounce = 60ms)

HS-CAN
30ms < t = 60ms

Table 3-42 Class B

3.11.1.6.3 Class C

Messages with a medium cycle time for HS-CAN. No or uncritical event triggered messages (tbounce > 60ms)

HS-CAN
60ms < t = 150ms

Table 3-43 Class C

3.11.1.6.4 Class D

Slow messages, cyclic and event triggered on HS-CAN

HS-CAN
150ms < t = 750ms

Table 3-44 Class D

3.11.1.6.5 Class E

Event triggered infrequent messages with a minimal repeat rate beyond class D

HS-CAN
t>750ms

Table 3-45 Class E

3.11.1.7 CAN Identifier Scheme

There are different messages types for different types of data (effective application, network management, diagnostic, transport protocol using by Multimedia and/or Telematic, etc.)

Range-ID	CAN- Messages Identifiers range		Message group Priority	Description
	Start/Base	End		
CANID_R00	0x000	0x07F	0	Reserved, don't use for application purpose
CANID_R01	0x080	0x0FF	0	very high priority application messages
CANID_R02	0x100	0x1FF	1	high priority application messages
CANID_R03	0x200	0x2FF	2	medium priority application messages
CANID_R04	0x300	0x3FF	3	normal priority application messages
CANID_R05	0x400	0x47F	4	OEM specific for multimedia- and/or telematic- functions

CANID_R05	0x480	0x4FF	4	OEM specific or OSEK- Network Management
CANID_R06	0x500	0x57F	5	OEM specific or AUTOSAR- Network Management
CANID_R06	0x580	0x5FF	5	Unsegmented messages transmission for special functions
CANID_R07	0x600	0x67F	6	OEM enhanced diagnostic (e.g. diagnostic response messages)
CANID_R07	0x680	0x6FF	6	OEM enhanced diagnostic (e.g. diagnostic request messages using TP with normal addressing format or special TP-connections)
CANID_R08	0x700	0x77F	7	Segmented messages transmission for special functions
CANID_R08	0x780	0x7DE	7	User specific(Proprietary area), measurement and calibration
CANID_R09	0x7DF	0x7EF	7	Legislated OBD/EOBD diagnostic
CANID_R10	0x7F0	0x7FF	7	Reserved range for further extensions, Do not use!

Table 3-46 CAN Identifier Classes

3.11.1.7.1 Messages-ID range for NM, services and diagnostics

COM_CANID_002:

These ID range are reserved for Network management and diagnostic purpose and shall not be used for anything else.

3.11.1.7.2 Proprietary Area

COM_CANID_003:

This area of CAN-identifier can be used for communication on sub-networks. On system-networks these identifiers are only allowed during development. Usage shall be clarified with the functional department of OEM.

3.11.1.7.3 Messages for Measuring Purposes

COM_CANID_004:

These messages are reserved for a supplier specific communication from and to it's ECU (e.g. CCP/XCP) and shall not be used for communication with other ECUs in any case. These Identifiers may be used only for development purposes and therefore are not allowed in series vehicles. In development vehicles these messages shall be sent on the bus only if requested from a measurement or calibration tool and shall not disturb the regular CAN communication. Hence CAN-message- identifiers in this range shall not be observed on the bus if a measurement and calibration tool like CANape or INCA is not in

use.

3.11.1.8 Message Transmit Types

COM_MSG_001:

Message transmit type describes when, or under which, conditions a message shall be transmitted. A network node is allowed to send a message at any time, but in general only two causes for a message transmission occur: A message may either be sent periodically using a fixed time interval, or the transmission may be triggered by some event.

3.11.1.8.1 Cyclic Messages

A "Cyclic" message shall be transmitted on a regular periodic schedule or fixed time, independently of changes in the signals contained within the message. The exact time between two transmissions, meaning the jitter of a cyclic message, may vary depending on the bus load.

Value	Minimum	Nominal	Maximum
t_{cycle}	10ms	-	-
Timeout factor	2	5	10

Table 3-47 Cyclic Message Properties

A node that receives a cyclic message shall observe the cycle time. If a cyclic message is not received within a certain amount of time defined as timeout, the receiving node shall take appropriate actions according to its specification document. The timeout for a cyclic message shall be defined globally in the vehicle communication matrix (e.g., 5 times the cycle time of the message).

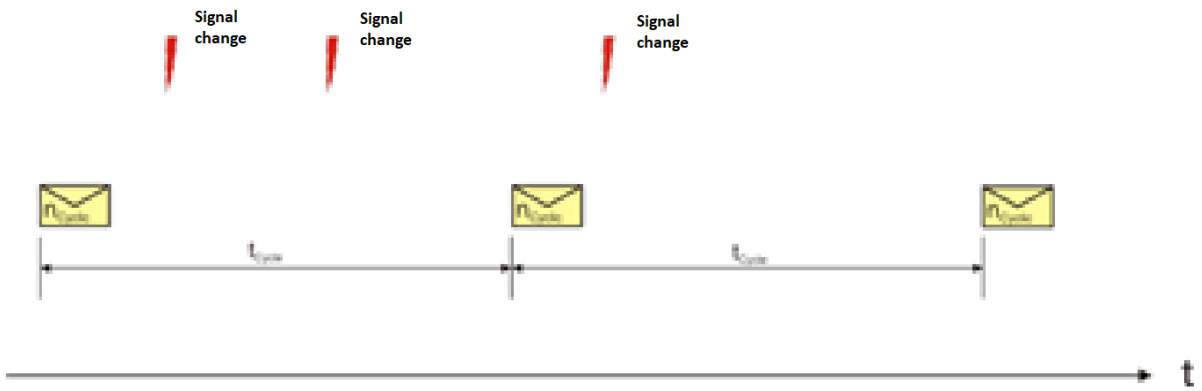


Figure 3-18 Timing Diagram Cyclic Message

The average cycle time of every cyclic message shall be within ± 2% of its defined cycle time. The maximum and minimum deviation from the specified cycle time depends on the value of the cycle time.

The maximum and minimum deviation from the cycle time shall not exceed the tolerance bands depicted in Figure: Range of cycle time tolerance.

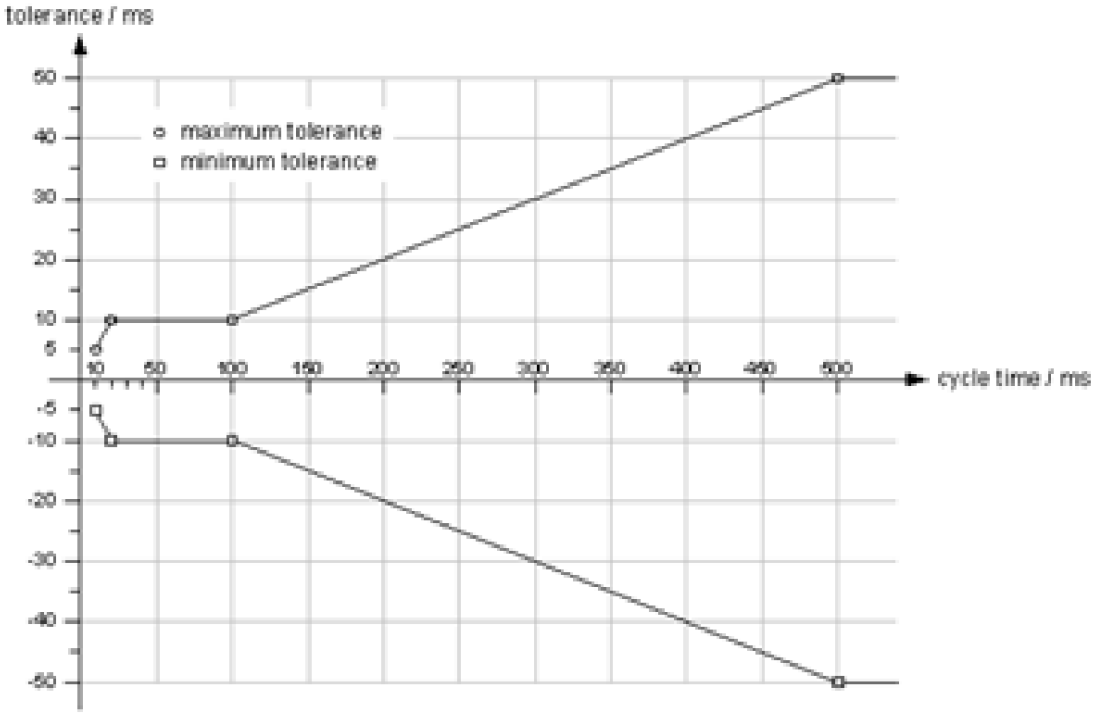


Figure 3-19 Range of Cycle Time Tolerance

3.11.1.8.2 Cyclic Active

A "Cyclic Active" message will be sent as long as the active requirement is fulfilled (e.g. button is pressed).

3.11.1.8.3 Cyclic Active without Stop Message

If the reason of activation is not present anymore, the cyclic transmission of the message stops. In Figure: Timing diagram cyclic active message, the message is sent only during t_{Active}.

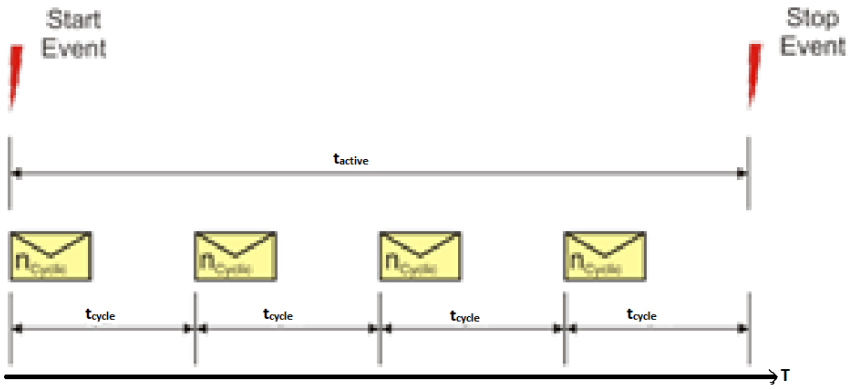


Figure 3-20 Timing Diagram Cyclic Active Message

3.11.1.8.4 Cyclic Active with Stop Message

After the end of the cyclic transmission a stop message shall be sent three times with a period e.g. t_{Rep}=100ms (Details specified in CAN-DBCs).

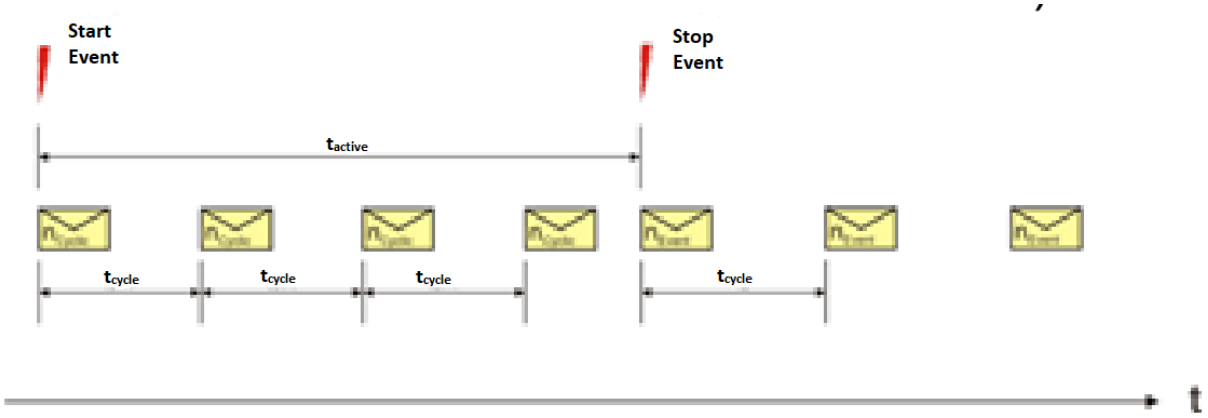


Figure 3-21 Timing Diagram Cyclic Active with Stop Message

3.11.1.8.5 Request Messages

Request messages are transmitted after request from a service message of another ECU. If the answer is not received within t_{wait}, the ECU will request the message again.

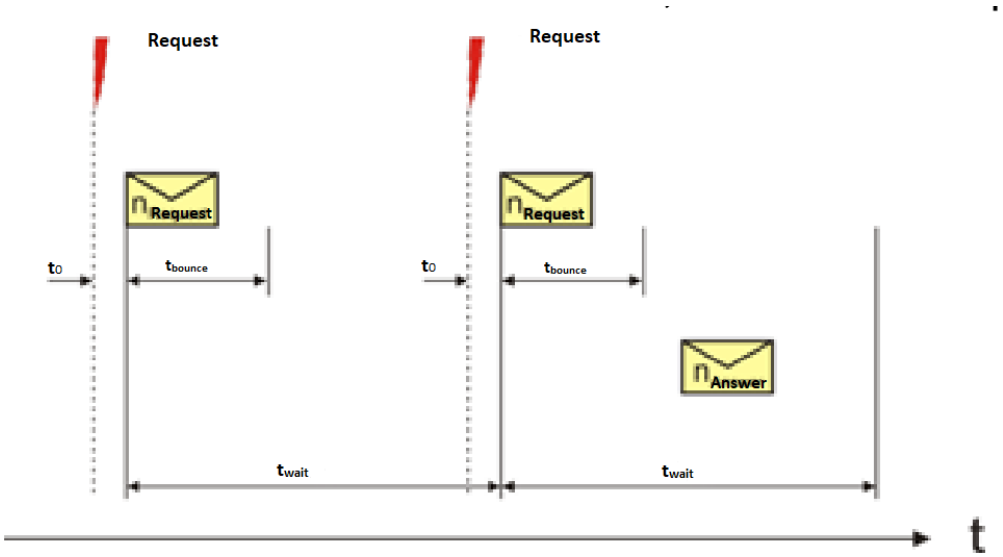


Figure 3-22 Timing Diagram Request Message

3.11.1.8.6 Event Triggered Messages

Event triggered messages are only transmitted on the bus if the according event occurs.

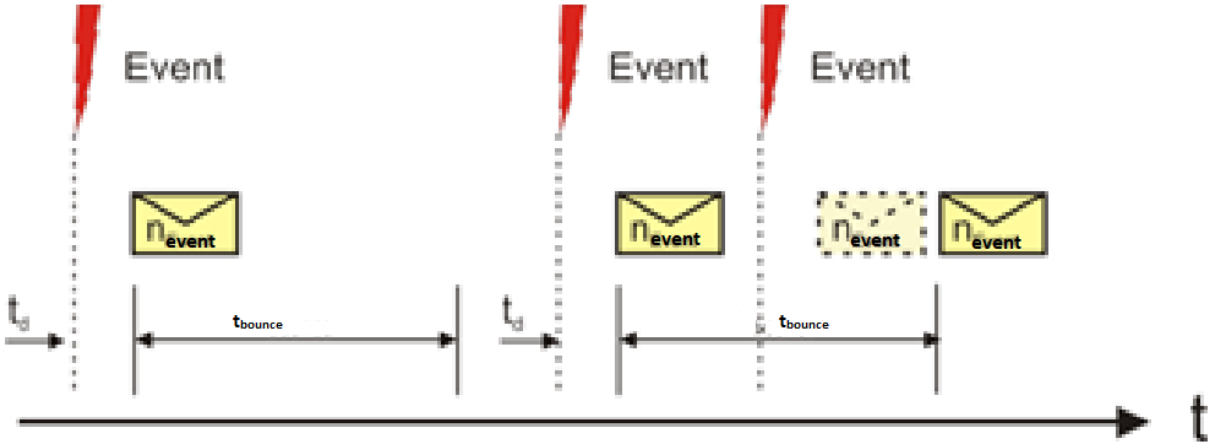


Figure 3-23 Timing Diagram Event Triggered Message

If a second event occurs within t_{bounce} of the first event the according message shall be sent after t_{bounce} of the first message has expired. The minimal time t_{bounce} between two consecutive transmissions of an event triggered message shall be defined globally in the vehicle communication matrix. The tolerance of t_{bounce} is given in Table: Tolerances of event interval time t_{min}.

Value	Minimum	Nominal	Maximum
t_{bounce}	4ms	5ms	10ms

Table 3-48 Tolerances of event interval time tmin

3.11.1.8.6.1 Event Triggered with Repetition

In that special case after the message, triggered by an event, is sent, the message shall be sent another e.g. 2-times with a period of e.g. t_{Rep}=100ms due to security reasons (Details specified in CAN-DBCs).

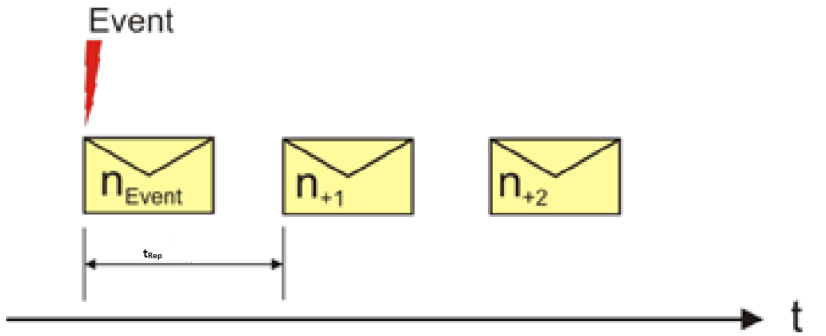


Figure 3-24 Timing Diagram Event Triggered Message with Repetition

3.11.1.8.6.2 Cyclic with Repetition on Event

In that special case after the message, triggered by an event, is sent, the message shall be sent another e.g. 2-times with a period of t_{Rep}=100ms due to security reasons (Details specified in CAN-DBCs). Additionally, to that the message shall be sent cyclic with cycle time t_{Cyclic} which is usually high (e.g. 1000ms).

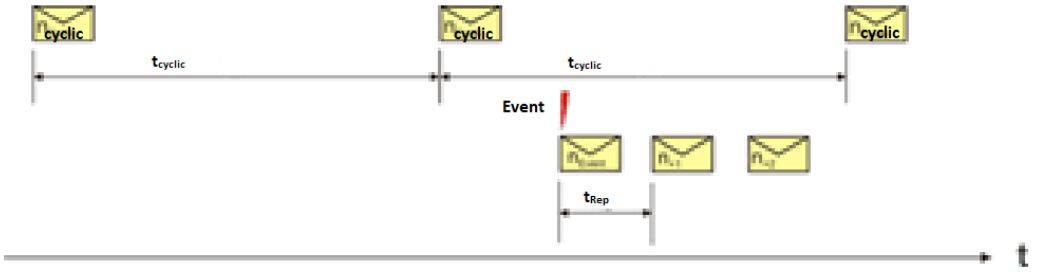


Figure 3-25 Timing Diagram Event Triggered with Repetition and Cyclic Message

3.11.1.9 Protection of Safety Messages Across Network

Safety-related data exchange shall be protected at runtime against the effects of faults within the communication link (e.g. from ECU-X on Bus-A trough Gateway to ECU-Y on Bus-B). Examples for such faults are random HW faults (e.g. corrupt registers of a CAN Controller), interference (e.g. due to EMC), and systematic faults within the software implementing the communication stack (CAN-driver and other communication software components). In order to ensure freedom from interference by software partitioning for example, measures to detect the following communication faults are taken into account (see NS 9 (ISO 26262) and NS 4 (AUTOSAR-specification).

- loss of peer to peer communication;
- unintended message repetition due to the same message being unintentionally sent again;
- message loss during transmission;
- insertion of messages due to receiver unintentionally receiving an additional message, which is interpreted to have correct source and destination addresses;
- sequencing due to the order of the data being changed during transmission, i.e. the data is not received in the same order as in which it was been sent
- message corruption due to one or more data bits in the message being changed during transmission;
- age delay due to the message being received correctly, but not in time
- blocking access to data bus due to a faulty node not adhering to the expected patterns of use and making excessive demands for service, thereby reducing its availability to other nodes, e.g. while wrongly waiting for non existing data

Depending on the security requirements of the message which should be protected, there are two methods, which can be combined in any order (checksum and/or alive counter).

COM_E2EALIVECRC_001:

ECU, which implement one or both mechanisms for protection of message content, shall provide a possibility (e.g. on diagnostic service request) to disable and re- enable checking of CRC and/or Alive counter. This is necessary to avoid many CRC- or Alive counter failures in the early FRSs.

3.11.1.9.1 Alive Counter

COM_E2EALIVE_001:

The actuality of the information which should be protected shall be ensured by the sender. Every sent message should increase the alive counter according to specification (0-14). If the value of the alive counter of a message did not change since the last transmission for a configurable number (e.g. 2-5 times, have to be defined separately at each receiver), the message shall be considered as invalid (Default values have to be defined at each receiver). Because alive counter is used to check only the actuality of a message, it will be not necessary to check, if the counter increments or decrements in a correct order (i.e. if it in-/de-crements by 1). The ECU application shall only check, if the value of the alive counter in n (configurable number e.g. 2-5) consecutive received message changes. This indicate, that the content of message is actualized or not. The sender of the messages will increment this counter by one each time the message is sent. The receiving ECU checks the actuality of the signals by subtracting the last received value Nlast from the new received value Nnew . The actuality of the information which should be protected shall be ensured by the sender. So if the result dN = (Nnew - Nlast) obtained is equal to zero(dN = 0), this means that the content of the last received message is not actualized by the sender. The maximum value of the counter value shall be set to 14. After counter value 14 counting shall start at zero again.

3.11.1.9.2 Application CRC

COM_E2ECRC_001:

The application CRC shall be calculated according to the specification of NS 10 (SAE J1850) CRC- 8-algoritm, which is typically quite efficient and may use hardware support. CRC-8 in NS 10 (SAE J1850) requires that the initial value and XOR value for CRC calculation is 0xFF.

3.11.1.9.2.1 Sample C Code of CRC-8 implementation according to SAE J1850

```
#define CRC8_TABLELENGTH 256u
#define UINT8_MAX ((uint8) 0xFF )
/* CRC8 according to SAE J1850 */
static const uint8 CRC8_J1850_TABLE[CRC8_TABLELENGTH] =
{
0x00 , 0x1D , 0x3A , 0x27 , 0x74 , 0x69 , 0x4E , 0x53 ,
0xE8 , 0xF5 , 0xD2 , 0xCF , 0x9C , 0x81 , 0xA6 , 0xBB ,
0xCD , 0xD0 , 0xF7 , 0xEA , 0xB9 , 0xA4 , 0x83 , 0x9E ,
0x25 , 0x38 , 0x1F , 0x02 , 0x51 , 0x4C , 0x6B , 0x76 ,
0x87 , 0x9A , 0xBD , 0xA0 , 0xF3 , 0xEE , 0xC9 , 0xD4 ,
0x6F , 0x72 , 0x55 , 0x48 , 0x1B , 0x06 , 0x21 , 0x3C ,
0x4A , 0x57 , 0x70 , 0x6D , 0x3E , 0x23 , 0x04 , 0x19 ,
0xA2 , 0xBF , 0x98 , 0x85 , 0xD6 , 0xCB , 0xEC , 0xF1 ,
0x13 , 0x0E , 0x29 , 0x34 , 0x67 , 0x7A , 0x5D , 0x40 ,
0xFB , 0xE6 , 0xC1 , 0xDC , 0x8F , 0x92 , 0xB5 , 0xA8 ,
0xDE , 0xC3 , 0xE4 , 0xF9 , 0xAA , 0xB7 , 0x90 , 0x8D ,
0x36 , 0x2B , 0x0C , 0x11 , 0x42 , 0x5F , 0x78 , 0x65 ,
```

```
0x94 , 0x89 , 0xAE , 0xB3 , 0xE0 , 0xFD , 0xDA , 0xC7 ,
0x7C , 0x61 , 0x46 , 0x5B , 0x08 , 0x15 , 0x32 , 0x2F ,
0x59 , 0x44 , 0x63 , 0x7E , 0x2D , 0x30 , 0x17 , 0x0A ,
0xB1 , 0xAC , 0x8B , 0x96 , 0xC5 , 0xD8 , 0xFF , 0xE2 ,
0x26 , 0x3B , 0x1C , 0x01 , 0x52 , 0x4F , 0x68 , 0x75 ,
0xCE , 0xD3 , 0xF4 , 0xE9 , 0xBA , 0xA7 , 0x80 , 0x9D ,
0xEB , 0xF6 , 0xD1 , 0xCC , 0x9F , 0x82 , 0xA5 , 0xB8 ,
0x03 , 0x1E , 0x39 , 0x24 , 0x77 , 0x6A , 0x4D , 0x50 ,
0xA1 , 0xBC , 0x9B , 0x86 , 0xD5 , 0xC8 , 0xEF , 0xF2 ,
0x49 , 0x54 , 0x73 , 0x6E , 0x3D , 0x20 , 0x07 , 0x1A ,
0x6C , 0x71 , 0x56 , 0x4B , 0x18 , 0x05 , 0x22 , 0x3F ,
0x84 , 0x99 , 0xBE , 0xA3 , 0xF0 , 0xED , 0xCA , 0xD7 ,
0x35 , 0x28 , 0x0F , 0x12 , 0x41 , 0x5C , 0x7B , 0x66 ,
0xDD , 0xC0 , 0xE7 , 0xFA , 0xA9 , 0xB4 , 0x93 , 0x8E ,
0xF8 , 0xE5 , 0xC2 , 0xDF , 0x8C , 0x91 , 0xB6 , 0xAB ,
0x10 , 0x0D , 0x2A , 0x37 , 0x64 , 0x79 , 0x5E , 0x43 ,
0xB2 , 0xAF , 0x88 , 0x95 , 0xC6 , 0xDB , 0xFC , 0xE1 ,
0x5A , 0x47 , 0x60 , 0x7D , 0x2E , 0x33 , 0x14 , 0x09 ,
0x7F , 0x62 , 0x45 , 0x58 , 0x0B , 0x16 , 0x31 , 0x2C ,
0x97 , 0x8A , 0xAD , 0xB0 , 0xE3 , 0xFE , 0xD9 , 0xC4
} ;
/*
* FUNCTION: uint8 crclib_getCRC8(const uint8 *xData, uint16 xDataLen)
* *
PARAMETERS: xData: Pointer(address) of data that CRC8 to be calculated
* xDataLen: size of data (number of bytes)
*
* RETURN: 8 Bits CRC- value
* *
DESCRIPTION: this function calculates the CRC8 value of given data
* according to SAE J1850
*/
uint8 crclib_getCRC8( const uint8 * xData, uint16 xDataLen)
{
uint8 idx;
uint8 crc = UINT8_MAX;
/* calculate CRC8 conform to SAE J1850 */
for (idx = 0 ; idx < xDataLen; idx ++ ) {
crc = CRC8_J1850_TABLE[crc ^ (xData[idx])];
}
/* the calculate CRC value shall be XOR with TUINT8_MAX ! */
crc ^= UINT8_MAX;
return crc;
} /* End of crclib_getCRC8(...) */
```

3.11.1.9.3 Message Layout with End-to-End protection

COM_E2EALIVECRC_002:

According to requirements COM_E2EALIVE_001 and COM_E2ECRC_001, each message, which content is be protected with alive counter or/and CRC, shall define the two signals ALIVE_XXX and CRC_XXX. The suffix XXX in the signal name is equal to the name of Message. The byte 0 of the message shall be reserved for CRC and the less significant nibble (bit 0 … bit 3) of byte 1 shall be reserved for the alive counter.

Byte	7	6	5	4	3	2	1								0							
Bit	…						7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
Signal	other Signals…						**ALIVE**_XXX								**CRC**_XXX							

Table 3-49 Layout of Message with Content Protection

COM_E2EALIVECRC_003:

Even if both signals ALIVE_XXX="alive counter" and CRC_XXX="CRC8 value" are always defined in a message with protected content, both protection mechanisms can be used independently from each other. This mean, that the content of a message can be protected only with CRC without computation or checking of the alive counter. Protecting a message content with alive counter don't obligatory need computation or checking of CRC.

COM_E2EALIVE_002:

The size of the signal ALIVE_XXX is 4 bits, hence it can take values between (0x0=0) and (x0F=15) where the value 0xF=15 equal to SNV (see COM_SIGSNV_001) indicating, that the alive counter is invalid.

COM_E2ECRC_002:

The size of signal CRC_XXX is 8 bits, hence it can take values between (0x00=0) and (0xFF=255). There is no invalid value for invalidation of a CRC-signal embedded in a message with protected content as specified in COM_E2ECRC_001. The whole content of the protected message excluding the CRC (Byte 0) itself and beginning with byte 1 to byte DLC-1 shall be used in the CRC-calculation.

3.12 Communication Error Handling

Electrical circuit continuity malfunctions (e.g. circuit shorted low, circuit shorted high, and circuit open) are typically caused by a wiring or connector problem, whereby electrical circuit performance/rationality malfunctions are typically more representative of an actual component failure.

Network diagnostic is defined as a system level approach to address common failures across multiple ECUs for monitoring and verifying electrical circuit continuity and functional performance of network communication systems. Most automotive electrical/electronic system faults can be divided into two major categories, network and power distribution. Network systems can be classified into four major failure groups representing the integrity of the bus electrical system and its associated data. Power distribution failures can be classified into a single failure group indicating the status of the power supply

to an ECU and its associated components. Power distribution faults are mentioned in this specification document because they also can cause network failures in most cases. Failure Detection is defined as the ECU's ability to identify failure conditions that affect the performance of the ECU or any associated circuits, components, or systems.

3.12.1 Network Electrical Failures

Network Electrical failures identify an ECU internal network electrical error, a network circuit short fault condition or a network circuit open fault condition.

COM_ERRELECTRICAL_001:

Since the CAN network uses a 2-wire twisted pair implementation, a specific DTC shall be defined for each possible circuit short and circuit open condition.

Network Failure mode	Description	Implementation
General Bus Failure	Defines a general communication network hardware malfunction (may indicate any electrical circuit continuity failure condition). A DTC for this Failure case ("General Bus Failure") shall be stored if following conditions are fulfilled: ·"no Busoff-indication", · "no short circuit to ground or to Battery, · "no open circuit (CAN_high, Can_low)", · "no TX-messages sent for a defined time(e.g. 1000ms)" · "no RX-messages received for a defined time(e.g. 1000ms)"	mandatory
CAN High connected to CL31	Defines a signal value measured at the CAN High communication line that is at or near zero scale (short to ground condition).	optional - mandatory if transceiver is able to detect
CAN Low connected to CL31	Defines a signal value measured at the CAN Low communication line that is at or near zero scale (short to ground).	optional - mandatory if transceiver is able to detect
CAN High connected to CL30	Defines a signal value measured at the CAN High communication line that is at or near full scale (short to battery condition).	optional - mandatory if transceiver is able to detect
CAN Low connected to CL30	Defines a signal value measured at the CAN Low communication line that is at or near full scale (short to battery condition).	optional - mandatory if transceiver is able to detect
CAN High Circuit Open	Defines a signal value measured at the CAN High communication line that is inactive (no response or interrupted condition).	optional - mandatory if transceiver is able to detect
CAN Low Circuit Open	Defines a signal value measured at the CAN Low communication line that is inactive (no response or interrupted condition).	optional - mandatory if transceiver is able to detect

CAN High and CAN Low Shorted	Defines a signal value measured at both the CAN High and CAN Low communication lines that is at the same voltage potential (shorted together condition).	optional - mandatory if transceiver is able to detect
Single-Wire Mode	Defines the reception of messages received from one or more ECU(s) on only a single wire, even though both wires seem to be functioning correctly. Single w ire operation is in general possible for low speed CAN-busses (up to 125 Kbps) and only if the used CAN-transceiver is fault tolerant.	optional - mandatory if transceiver is able to detect

Table 3-50 Network Electrical Failures

3.12.1.1 Network Bus-Off Failure

Condition where all bus message transmission has been disrupted and the ECU is logically disabled from the communication network to prevent further bus behavioral disruptions and potential permanent damage to the network.

COM_ERRBUSOFF_001:

Every ECU shall be able to detect and report Bus-Off failure as describe in following (more details on bus off mechanism can be founded in NS 1 (ISO 11898-1) , NS 2 (ISO 11898-2), NS 3 (ISO 11898-3) and section CAN_DataLink_Layer.

Network Failure mode	Description
Bus off	Defines the status information about failure conditions of the transceiver (e.g. Bus-Off) provided by the communication controller. This includes all failure modes detectable and reported by the bus controller which are not covered by any of the other failure modes and not related to the electrical/physical integrity of the bus.

Table 3-51 Network failure mode Bus-Off

3.12.1.2 Network Node Integration Failures

Network node integration failures identify whether or not an ECU that is designed for the vehicle is actually communicating on the network. Network node integration fault - detection improves both vehicle build confidence and network fault isolation. A Network node integration failure may indicate a miss-build condition, a configuration error, an electrical hardware failure condition, or an ECU internal failure.

COM_ERRNODEINTEGRA_001:

Every ECU shall be able to detect and report following node integration Failures in the communication network with exception of "Additional ECU Detected".

NOTE:

For detection of present nodes in the communication network, the AUTOSAR-NM provide the necessary functionality and API. Every ECU implementing AUTOSAR-NM can detect additional nodes which are

integrated in the Network, but it is not necessary to report an error.

Network Failure mode	Description
Lost Communication with ECU ‘X’	Defines the absence of specific bus messages as expected from ECU "X" at predefined periodic rates, or as required from ECU "X" for acknowledgement and/or functional purposes.
Additional ECU Detected	Defines the presence of an ECU on the network even though the ECU is not programmed to be on the vehicle based on the Vehicle Configuration.

Table 3-52 Network failure mode Integration failures

3.12.1.3 Message- and Signal- Content Failures

Message- and Signal- content failures are used to flag data integrity issues associated with a received CAN- message or signal. The Message- and Signal- content failure may be an intentional signal transmitted by the originating ECU or gateway ECUs rather than a network fault condition.

COM_ERRMSGSIG_001:

Every ECU shall be able to detect and report following message and signal- failures:

Network Failure mode	Description
Implausible Signal Received from ECU ‘X’	Defines a signal element received through a bus message with 'implausible or invalid data' indicating signal available but with loss of integrity (permanent condition due to component failure condition).
Implausible message content Received from ECU ‘X’	Defines a message content with 'implausible or invalid data' indicating signals available but with loss of integrity (permanent condition due to CRC- or alive counter - error detected).

Table 3-53 Network failure mode message-and signal-content failures

3.12.1.4 System Power Supply and Communication Error Detection

Power Distribution failures are used to flag when a supply voltage range exceeds specification.

Network Failure mode	Description
System Voltage to Low	Defines a signal value received as a bussed input which falls below the lower limit of the system voltage range.
System Voltage High	Defines a signal value received as a bussed input which exceeds the upper limit of the system voltage range.

Table 3-54 Network failure mode system power supply failures

COM_ERRCL30_001:
On detection of this two failures ("System Voltage High" or "System Voltage Low"), no communication failure as defined in section “Network Bus-Off Failure” and “Message and signal content failures” shall be reported or retained even, if they are detected during system voltage failure.

3.12.1.5 Ignition State and Communication Error Detection

Network Communication failure detection is performed only with the ignition in the "ON" state. The ignition state (CL15) is used as enable condition for Network Communication failure detection. As not all ECUs are battery fed, these ECUs for example connected on HS-CAN will not transmit messages when the ignition state does not equal "ON".

NOTE:

The bussed input requirements, as specified in section CAN- Communication, are used with the following enable condition requirements.

COM_CL15_001:

The ignition state (CL15) received over the bus shall take precedence over direct ignition state measurement to achieve standardized ignition state fault handling for all ECUs in the OEM-vehicle.

COM_CL15_002:

If the bus communication has failed or an invalid ignition state signal is received, and a locally measured ignition state is available, the locally measured ignition state shall be used.

COM_CL15_003:

If the bus communication has failed or an invalid ignition state signal is received, and a locally measured ignition state is not available, the last known valid Ignition state (CL15) shall be used for a configurable amount of time and then shall be considered as OFF in the current power cycle, if the above denoted situation persists.

COM_CL15_004:

If the bus communication has failed and a valid ignition state signal has not been received within the

current power cycle, and a locally measured ignition state value is not available or valid, the ECU shall assume that the Ignition state is OFF and therefore the enable condition is not fulfilled for Network Communication failure detection.

4 Development Environments

Not relevant.

4.1 Development Environment

N/A

5 References

5.1 Quoted Documents

Ref. No.	Name	Version
QD1	Basic_Diagnostic_Specification	latest
QD2	ECU_Identifier_Specification	latest
QD3	Standard_Component	latest
QD4	basic_diagnostic_System_specification	latest
QD5	Flash_Specification	latest
QD6	Coding_Specification	latest
QD7	Basic_DID_Definition_Specification	latest
QD8	Part 2 Body Control Module (BCM)	latest
QD9	System_schematics_VINFAST	latest
QD10	CES_General_Electric	latest
QD11	Part 2 Gateway (XGW)	latest

Table 5-1 Quoted documents

5.2 Norms and Standards

Ref. No.	Name	Version
NS1	ISO 11898-1, Road Vehicles -Controller area network (CAN) Part1 Data link layer an physical signaling.	01.12.2003
NS2	ISO 11898-2, Road Vehicles - Controller area Network (CAN) Part 2: High-speed medium access unit.	01.12.2003
NS3	ISO 11898-3, Road Vehicles - Controller area network (CAN) Part 3: Low-speed, fault tolerant- medium dependent interface	01.06.2006
NS4	AUTOSAR - Specification of Network Management Interface Document Identification No: 228 Part of Release: 4.3.1	08.12.2017
NS5	AUTOSAR – Specification of CAN Network Management Document Identification No: 013 Part of Release: 4.3.1	08.12.2017

NS6	ISO 15765-2 (2016-04-01), Road vehicles — Diagnostic communication over Controller Area Networks (DoCAN) — Part 2: Transport protocol and network layer services	01.04.2016
NS7	ISO 15765-3, Road vehicles - Diagnostic on Controlled Area Networks (CAN) Part 3: Implementation of unified diagnostic services (UDS on CAN)	2004(E), 15.10.2004
NS8	ISO/IEC 8859-1 Information technology -- 8-bit single-byte coded graphic character sets -- Part 1: Latin alphabet No. 1	1998
NS9	ISO 26262 Road vehicles – Functional safety	01.08.2012
NS10	SAE J1850 Verification Test procedures	05.06.2006

Table 5-2 List of references